

FALCONSTOR®

# FALCONSTOR STORSAFE

IBM POWER DEPLOYMENT GUIDE

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## ***FalconStor® StorSafe® for IBM Power Deployment Guide***

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# Introduction

FalconStor StorSafe is an optimized backup and deduplication solution that provides virtual tape library emulation, Network-Attached Storage (NAS), high-speed backup/restore, data archival to supported S3 clouds for long-term storage, global data deduplication, and enterprise-wide replication, without requiring changes to the existing environment.

FalconStor StorSight is a single integrated platform that simplifies the management of data across legacy, modern, and virtual storage environments. StorSight gathers and consolidates information coming from different StorSafe servers into a scalable repository of services, tenants, users, predictive analytics, alert rules, reporting, and historical data. StorSight provides a web-based portal for centralized management and monitoring of multiple backup and deduplication servers.

This guide describes how to create an IBM Power Virtual Server (PowerVS) running FalconStor StorSafe in IBM Public or Private Cloud and how to configure FalconStor StorSafe using FalconStor StorSight.

## Intended audience

This guide is intended for the following individuals deploying the FalconStor StorSafe solution:

- Storage architects
- Consultants
- System Administrators

Individuals performing the deployment should have strong experience with the following products and technologies:

- IBM Cloud
- FalconStor StorSafe and StorSight solution
- Network design, configuration, and security
- Ethernet topologies, network routers, VPN, VLAN
- iSCSI technologies
- Fibre Channel SAN technologies (only if FC is used on-premises)

## Resources and helpful links

| Resource                                                                                           | Link                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FalconStor Technical Support                                                                       | <a href="https://www.falconstor.com/support/technical-support">https://www.falconstor.com/support/technical-support</a>                                                                                                                                                                                         |
| FalconStor Sizing Calculator                                                                       | <a href="http://ibmsizing.falconstor.com">http://ibmsizing.falconstor.com</a>                                                                                                                                                                                                                                   |
| FalconStor Certification Matrix                                                                    | <a href="https://www.falconstor.com/support/certification-matrix">https://www.falconstor.com/support/certification-matrix</a>                                                                                                                                                                                   |
| FalconStor license registration email                                                              | <a href="mailto:activate.keycode@falconstor.com">activate.keycode@falconstor.com</a>                                                                                                                                                                                                                            |
| FalconStor StorSight Virtual Hard Disk (VHD) image for x-86 servers in Virtual Private Cloud (VPC) | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-vpc.vhd.zip">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-vpc.vhd.zip</a>                                                                         |
| FalconStor StorSight VHD image for x-86 servers in Classic infrastructure                          | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-classic.vhd.zip">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-classic.vhd.zip</a>                                                                 |
| FalconStor StorSight SAK image for x-86 servers                                                    | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-sak.zip">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor-StorSight-sak.zip</a>                                                                                 |
| FalconStor StorSafe SAK image for on-premises Power servers                                        | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/SAK-Install-8.8.x-StorSafe-11.13-12328-ppc64le-Linux8.iso">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/SAK-Install-8.8.x-StorSafe-11.13-12328-ppc64le-Linux8.iso</a>                       |
| FalconStor SAK Image Installation Guide                                                            | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20Server%20SAK%20ISO%20Image%20Installation%20Guide.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20Server%20SAK%20ISO%20Image%20Installation%20Guide.pdf</a>     |
| FalconStor Server Virtual Appliance Installation Guide                                             | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20Server%20Virtual%20Appliance%20Installation%20Guide.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20Server%20Virtual%20Appliance%20Installation%20Guide.pdf</a> |
| FalconStor StorSafe with StorSight Release Notes                                                   | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20StorSafe%20with%20StorSight%20Release%20Notes.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20StorSafe%20with%20StorSight%20Release%20Notes.pdf</a>             |
| FalconStor StorSafe with StorSight User Guide                                                      | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20StorSafe%20with%20StorSight%20User%20Guide.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20StorSafe%20with%20StorSight%20User%20Guide.pdf</a>                   |
| FalconStor StorSafe Use Case - Veeam and AIX                                                       | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20Use%20Case%20-%20Veeam%20and%20AIX.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20Use%20Case%20-%20Veeam%20and%20AIX.pdf</a>                                       |
| FalconStor StorSafe Use Case – IBM Spectrum Protect and AIX                                        | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20Use%20Case%20-%20IBM%20SP%20and%20AIX.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20Use%20Case%20-%20IBM%20SP%20and%20AIX.pdf</a>                                 |

|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FalconStor StorSafe NAS Use Case - SAP HANA                               | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20NAS%20Use%20Case%20-%20SAP%20HANA.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20NAS%20Use%20Case%20-%20SAP%20HANA.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| FalconStor StorSafe NAS Use Case - Oracle RMAN                            | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20NAS%20Use%20Case%20-%20RMAN.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/StorSafe%20NAS%20Use%20Case%20-%20RMAN.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| FalconStor VTL 10.03 to StorSafe 11.13 Upgrade Patch                      | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/UpgradeVTL10.03ppc64letoStorSafe11.13Tile">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/UpgradeVTL10.03ppc64letoStorSafe11.13Tile</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| FalconStor VTL 10.03 to StorSafe 11.13 Upgrade Guide                      | <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20VTL%2010.03%20to%20StorSafe%2011.13%20Upgrade%20Guide.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/FalconStor%20VTL%2010.03%20to%20StorSafe%2011.13%20Upgrade%20Guide.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| IBM documentation for installation, configuration, and cloud connectivity | <p>IBM Power Systems Virtual Server Guide for IBM<br/> <a href="https://www.redbooks.ibm.com/Redbooks.nsf/RedpieceAbstracts/sg248513.html?Open">https://www.redbooks.ibm.com/Redbooks.nsf/RedpieceAbstracts/sg248513.html?Open</a></p> <p>IBM Power Virtual Server Virtual Private Network Connectivity<br/> <a href="https://cloud.ibm.com/media/docs/downloads/power-iaas-tutorials/PowerVS_VPN_Tutorial_v1.pdf">https://cloud.ibm.com/media/docs/downloads/power-iaas-tutorials/PowerVS_VPN_Tutorial_v1.pdf</a></p> <p>Modifying a Power Systems Virtual Server instance<br/> <a href="https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-modifying-server">https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-modifying-server</a></p>                                                                               |
| IBM documentation for IBM i Backups with IBM Power Virtual Server         | <p>IBM i Backups with IBM Power Virtual Server Tutorial<br/> <a href="https://cloud.ibm.com/media/docs/downloads/power-iaas-tutorials/PowerVS_IBMi_Backups_Tutorial_v1.pdf">https://cloud.ibm.com/media/docs/downloads/power-iaas-tutorials/PowerVS_IBMi_Backups_Tutorial_v1.pdf</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| IBM documentation for IBM i Migration to IBM Power Virtual Server         | <p>IBM i Migration to IBM Power Virtual Server using FalconStor<br/> <a href="https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/IBM%20i%20Migration%20to%20IBM%20Power%20Virtual%20Server%20using%20FalconStor.pdf">https://falconstor-download.s3.us-east.cloud-object-storage.appdomain.cloud/IBM%20i%20Migration%20to%20IBM%20Power%20Virtual%20Server%20using%20FalconStor.pdf</a></p>                                                                                                                                                                                                                                                                                                                                                                                                             |
| IBM documentation for iSCSI clients                                       | <p>IBM i Support for Attaching an iSCSI VTL<br/> <a href="#">IBM i Support for iSCSI VTL 1.6_3.pdf</a></p> <p>IBM i 7.5 iSCSI Boot<br/> <a href="https://download4.boulder.ibm.com/sar/CMA/HMA/0b11y/3/MF70433.readme.html?_gl=1*m2mt0*_ga*MjExODMzNzI4NC4xNTc0NjQwNjE5*_ga_FYECCCS21D*MTY5NjI0OTMxNC40MTcuMS4xNjk2MjUxOTUzLjAuMC4w">https://download4.boulder.ibm.com/sar/CMA/HMA/0b11y/3/MF70433.readme.html?_gl=1*m2mt0*_ga*MjExODMzNzI4NC4xNTc0NjQwNjE5*_ga_FYECCCS21D*MTY5NjI0OTMxNC40MTcuMS4xNjk2MjUxOTUzLjAuMC4w</a><br/> <a href="https://www.ibm.com/docs/en/linux-on-z?topic=server-boot-process">https://www.ibm.com/docs/en/linux-on-z?topic=server-boot-process</a></p>                                                                                                                                             |
| IBM documentation for Cloud Connections, Direct Link, and Proxy Server    | <p>Managing IBM Cloud connections<br/> <a href="https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-cloud-connections">https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-cloud-connections</a></p> <p>Using IBM Cloud Direct Link to connect to IBM Cloud Object Storage<br/> <a href="https://cloud.ibm.com/docs/direct-link?topic=direct-link-using-ibm-cloud-direct-link-to-connect-to-ibm-cloud-object-storage">https://cloud.ibm.com/docs/direct-link?topic=direct-link-using-ibm-cloud-direct-link-to-connect-to-ibm-cloud-object-storage</a></p> <p>Ordering Direct Link Connect for Power Systems Virtual Servers<br/> <a href="https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-ordering-direct-link-connect">https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-ordering-direct-link-connect</a></p> |

|                                             |                                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IBM documentation for DNS FAQ               | Defining Domain Name Servers (DNS)<br><a href="https://ondeck.console.cloud.ibm.com/docs/dns?topic=dns-dns-faq">https://ondeck.console.cloud.ibm.com/docs/dns?topic=dns-dns-faq</a> |
| IBM documentation for Cloud Transit Gateway | <a href="#">Ordering IBM Cloud Transit Gateway</a><br><a href="#">Using virtual private endpoints for VPC to privately connect to IBM Cloud Transit Gateway.</a>                    |

# Deployment Scenarios

The following scenarios are examples of different deployments for FalconStor StorSafe with IBM Cloud:

- Native backup in the cloud: StorSafe in IBM Cloud serves different IBM i host clients for backup.
- IBM cloud-to-cloud replication: StorSafe in IBM Cloud replicates data to another StorSafe.
- Hybrid Cloud: On-premises StorSafe replicates data to a StorSafe in IBM Cloud.
- Workload migration: On-premises StorSafe replicates data to a StorSafe in IBM Cloud, migrates backup client workload, and then removes the on-premises StorSafe.

## Connectivity options

The network infrastructure can be very different for each customer, depending on their deployment use case, security, performance, and reliability requirements. Consult IBM networking references as additional connection options may become available for IBM PowerVS. This section is meant for general considerations and provides some information and screenshot samples as guidelines.

To configure connectivity between the components, you can use public IP addresses for cloud server access, but for higher security, it is better to use private IP addresses and Virtual Private Networks (VPNs) for a secure connection between machines in the primary site and machines in the IBM Cloud.

You can order Direct Link (DL) Connect to allow your PowerVS to communicate with Linux/Window virtual servers in IBM Cloud and also with all other IBM Cloud services, such as Cloud Object Storage (COS) and VMware services. Alternatively, You can use a transit gateway router for communication between PowerVS and Virtual Private Cloud (VPC).

For servers at on-premises sites, you need a physical router; for servers in the cloud, you need a Virtual Router Appliance (VRA), such as Vyatta or Juniper, that allows IBM Cloud users to selectively route private network traffic through firewall and VPN features.

You can use Megaport, which enables direct and dedicated connectivity between the primary site and IBM Cloud, to overcome the use of the public internet. Megaport is a software layer to manage network connections, allowing private point-to-point connectivity between any of the locations on the Megaport global network infrastructure. You can use a service key in your Megaport account to create a Virtual Cross Connect (VXC) from a port on a Megaport Cloud Router (MCR) to a port on the primary site. The key creator has control over limiting the bandwidth of the connection and can also specify the Virtual Local Area Networks (VLAN) ID for a single-use key.

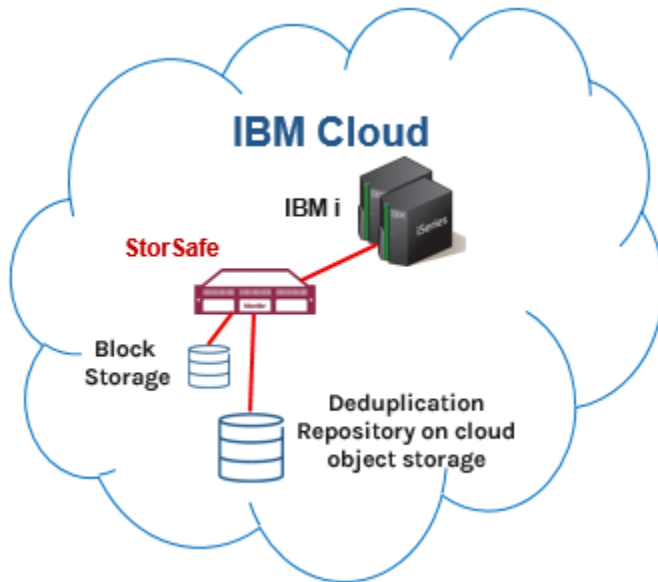
You can use IBM Deployable Architecture from the catalog to build a Power Systems Virtual Server workspace with VPC landing zone. Refer to the appendix in this guide for details.

You can make remote connections to IBM COS using different endpoints; refer to <https://cloud.ibm.com/docs/cloud-object-storage?topic=cloud-object-storage-endpoints>:

- **Public endpoints** can accept requests from anywhere and charges are assessed on outgoing bandwidth. Incoming bandwidth is free. In this case, you will also need to have a public IP for your server, which may be a security concern.
- **Private endpoints** are not available from a VPC, but are available for most requests originating from within IBM Cloud. Private endpoints do not incur charges for any outgoing or incoming bandwidth even if the traffic is cross regions or across data centers.
- **Direct endpoints** are used for requests originating from VPCs. Direct endpoints provide better performance over Public endpoints and do not incur charges for any outgoing or incoming bandwidth even if the traffic is cross regions or across data centers. Direct endpoints can be accessed through Virtual Private Endpoint gateways.

## Native Backup Deployment

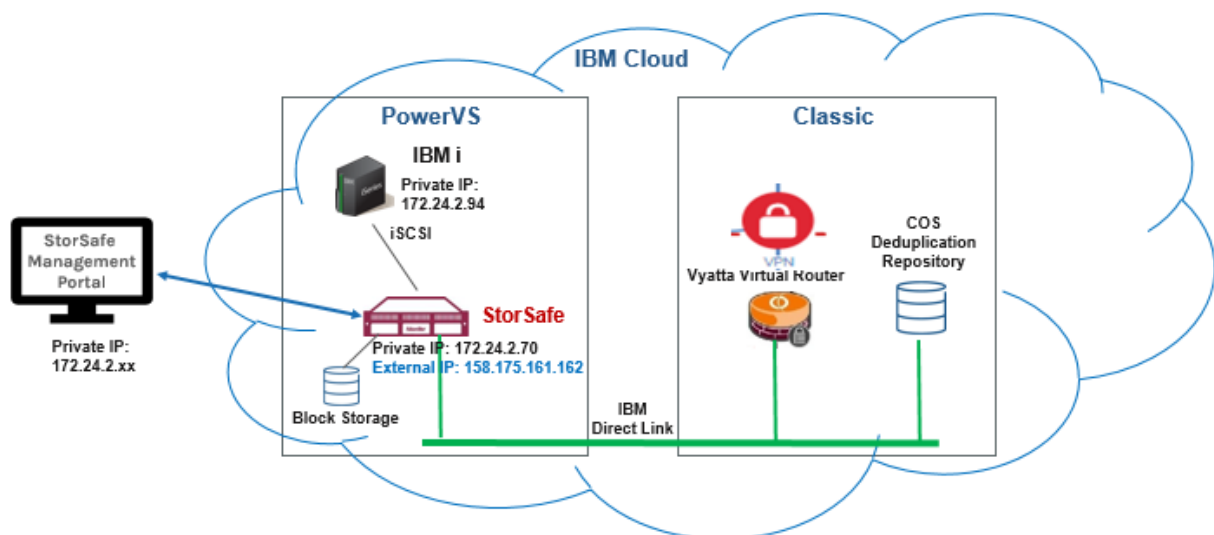
Several IBM i host clients are connected via iSCSI to StorSafe in the cloud to back up data via IBM Backup, Recovery & Media Services (BRMS) to virtual tape libraries. StorSafe uses IBM COS as the data devices for the deduplication repository via a Generic S3 object storage account.



## Connectivity

The following connections are required:

- Network connections between StorSafe and the StorSight management portal
- iSCSI connections between StorSafe and IBM i host clients
- IBM Cloud Direct Link connection or site VPN between StorSafe and the IBM COS
- Internet connection with the FalconStor license server for online registration (optional)



**Example**

Network default interface `eth0` has a public IP address and is used for all external connections.

Network interface `eth1` has a private IP address and is used to add routes to other networks.

| Public networks                                                                      |                 |                 |                 |                   |         |                    |
|--------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-------------------|---------|--------------------|
|  On |                 |                 |                 |                   |         |                    |
| Name                                                                                 | IP address      | External IP     | Gateway         | MAC address       | VLAN ID | CIDR               |
| public-192.168.150.160-29-VLAN_2016                                                  | 192.168.150.162 | 158.175.161.162 | 192.168.150.161 | fa:d3:db:c3:12:20 | 2016    | 192.168.150.160/29 |

| Private networks                    |             |             |                   |         |                                         |                                                                                                            |
|-------------------------------------|-------------|-------------|-------------------|---------|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| <input type="text" value="Search"/> |             |             |                   |         | <a href="#">Attach existing network</a> |                                                                                                            |
| Name                                | IP address  | Gateway     | MAC address       | VLAN ID | CIDR                                    |                                                                                                            |
| private-172.24.2.64/27-VLAN-661     | 172.24.2.70 | 172.24.2.65 | fa:d3:db:c3:12:21 | 661     | 172.24.2.64/27                          | <a href="#">Detach</a>  |

The system command displays the network configuration on StorSafe as follows:

```
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.150.162 netmask 255.255.255.248 broadcast
    192.168.150.167

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9000
    inet 172.24.2.70 netmask 255.255.255.224 broadcast 172.24.2.95
```

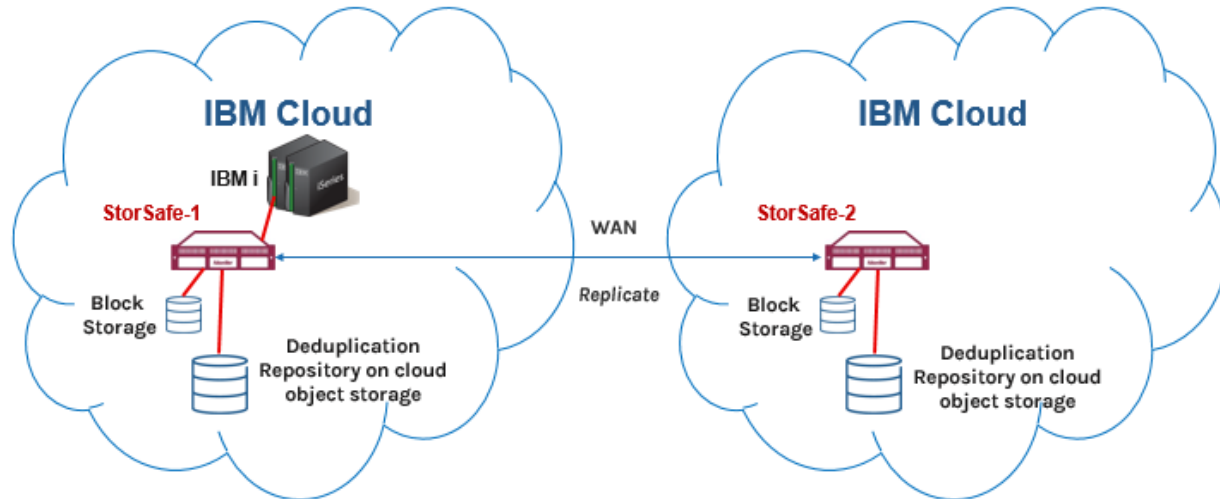
Create routes from the private IP to connect to the other networks. For example, `ADDRESS1=10.0.0.0` is for routing to the cloud infrastructure and `ADDRESS2=172.22.0.0` is added for routing to other private networks.

Any other customized networks to the route can be added using sequential numbers as a postfix.

```
# cat /etc/sysconfig/network-scripts/route-eth1
# Created by cloud-init on instance boot automatically, do not edit.
#
ADDRESS0=172.24.2.64
GATEWAY0=172.24.2.65
NETMASK0=255.255.255.224
ADDRESS1=10.0.0.0
GATEWAY1=172.24.2.65
NETMASK1=255.0.0.0
ADDRESS2=172.22.0.0
GATEWAY2=172.24.2.65
NETMASK2=255.255.0.0
```

## Cloud-to-Cloud Replication Deployment

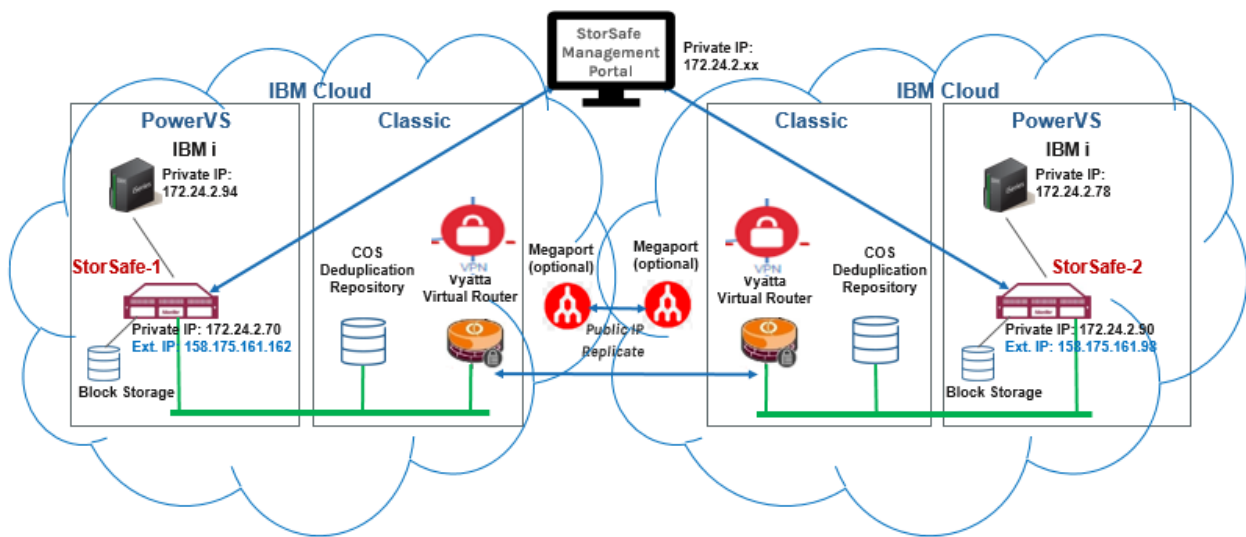
StorSafe in IBM Cloud replicates data over a WAN to another StorSafe in the cloud for data protection and disaster recovery. StorSafe source and replica servers use IBM COS as the data devices for the deduplication repository via Generic S3 object storage accounts.



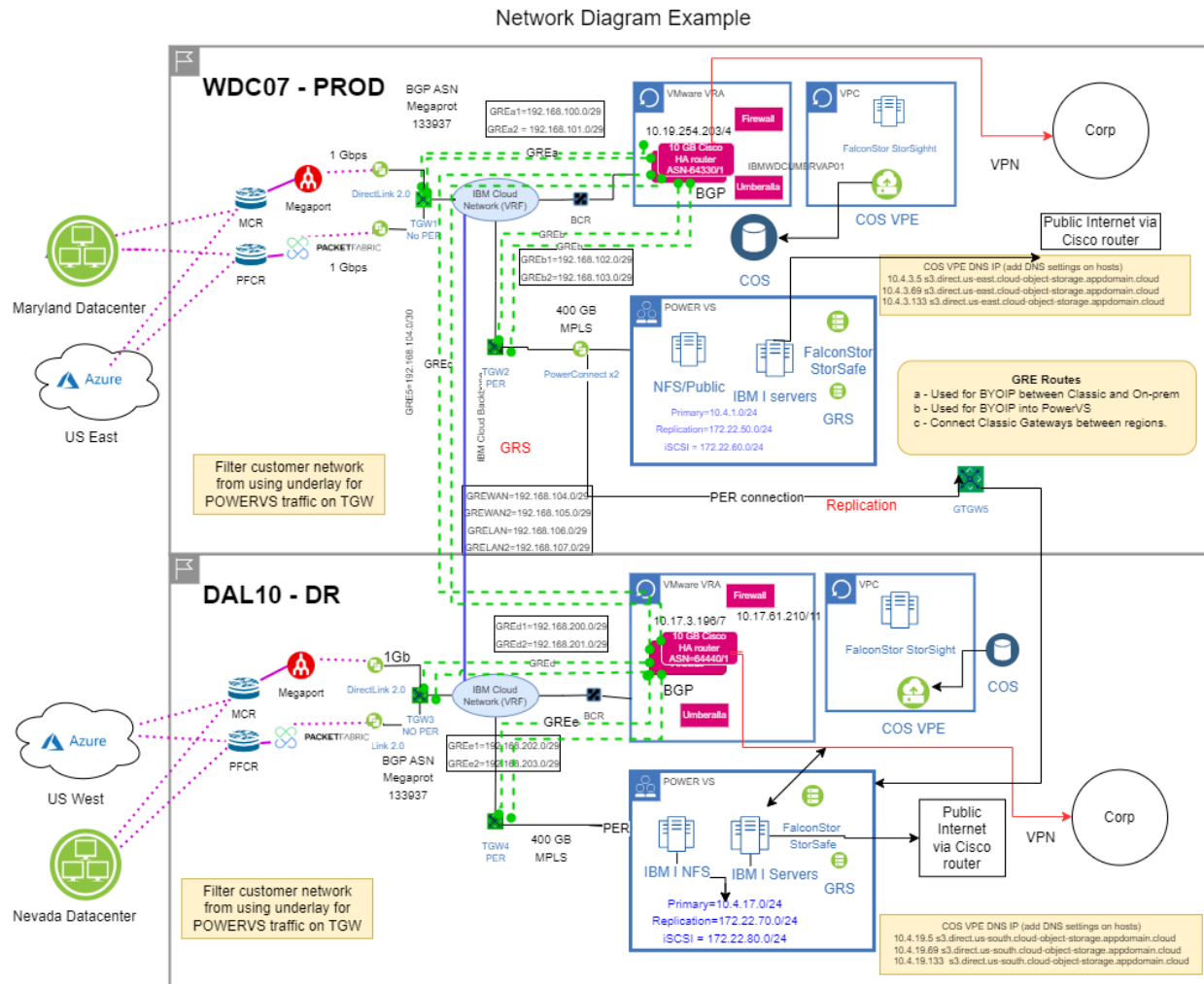
## Connectivity

The following connections are required:

- Network connections between StorSafe and the StorSight management portal
- Network connection between the source and the replica StorSafe servers by virtual routers or Megaport
- iSCSI connections between StorSafe and IBM i host clients
- IBM Cloud Direct Link connection or site VPN between StorSafe and the IBM COS
- Internet connection with the FalconStor license server for online registration (optional)

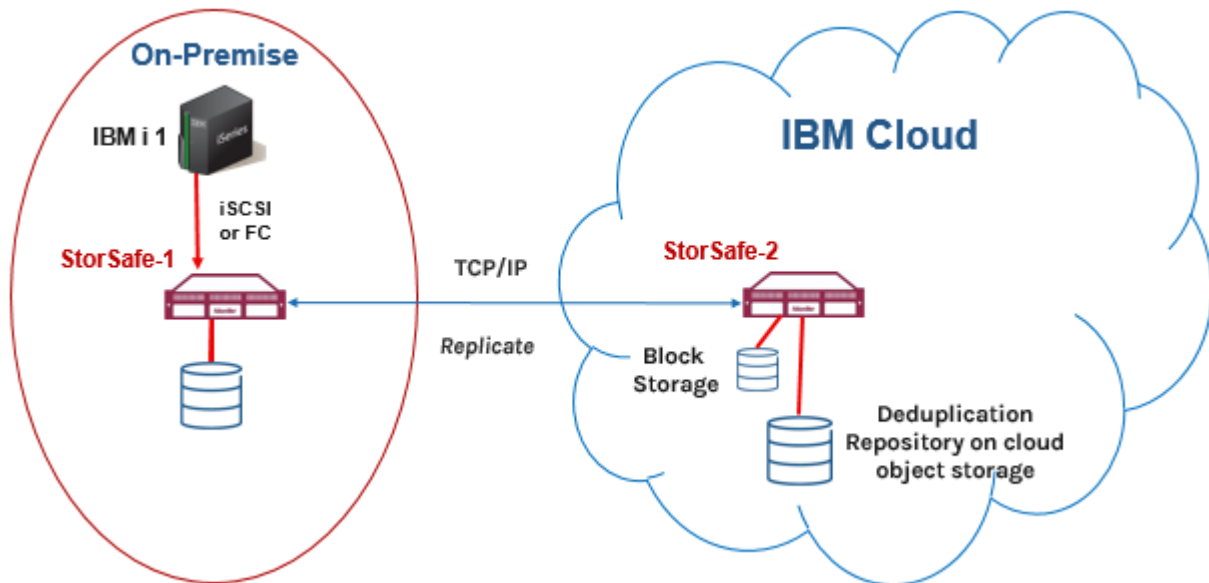


## Deployment Network Example



## Hybrid Cloud Deployment

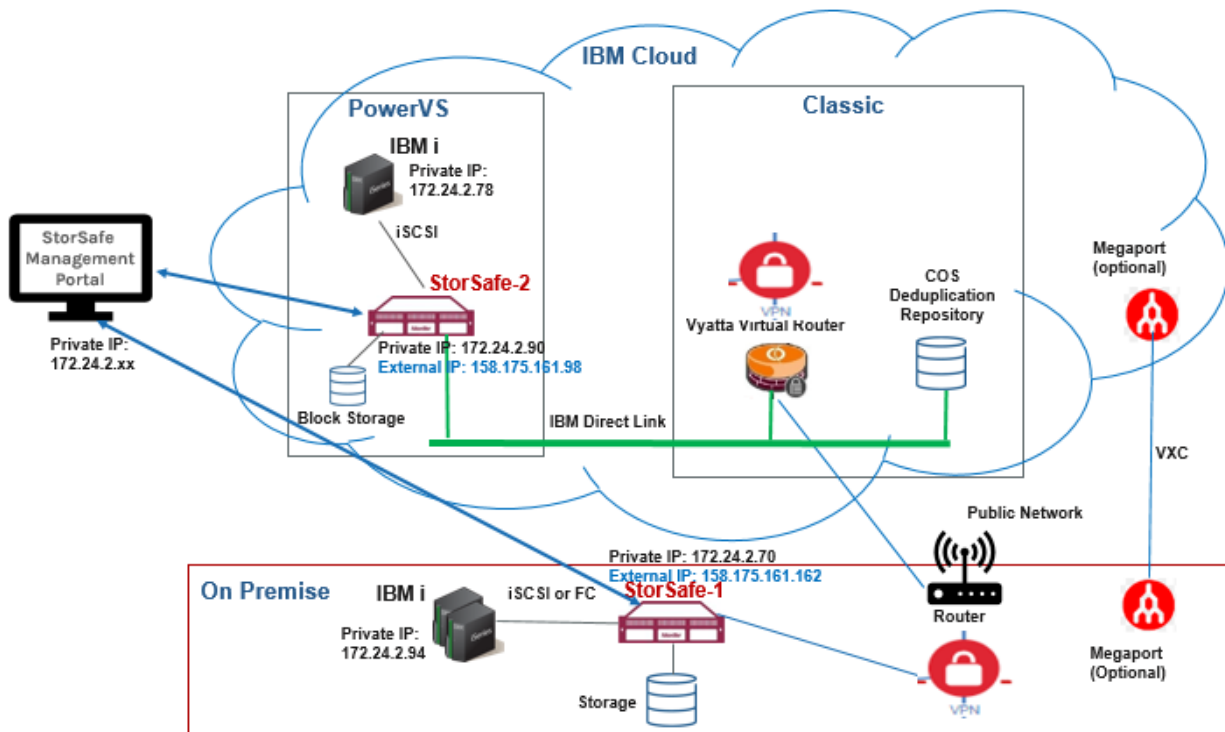
An on-premises StorSafe server deduplicates data and replicates to a StorSafe server in IBM Cloud for data protection and disaster recovery purposes. The on-premises StorSafe uses SCSI devices for the deduplication repository. The StorSafe target server in the cloud uses IBM COS as the data devices for the deduplication repository via a Generic S3 object storage account.



### Connectivity

The following connections are required:

- Network connection between the on-premises StorSafe and the PowerVS StorSafe in IBM Cloud
- Network connection between StorSight in the x86 infrastructure and the on-premises StorSafe
- Network connections between StorSight in the x86 infrastructure and the PowerVS StorSafe in IBM Cloud
- iSCSI or Fibre Channel connection between IBM i host clients and the on-premises StorSafe
- iSCSI connections between IBM i host clients and the PowerVS StorSafe in IBM Cloud
- IBM Cloud Direct Link connection or site VPN between the PowerVS StorSafe and the IBM COS
- Internet connection with the FalconStor license server for online registration



## Workload Migration Deployment

IBM i backup clients of on-premises StorSafe servers are to be moved to the cloud via StorSafe data replication. StorSafe in the cloud uses IBM COS as the data devices for the deduplication repository via a Generic S3 object storage account.

The following describes how IBM Cloud can be used for workload migration:

1. IBM BRMS running at the primary site on an IBM i host client performs I/O to a FalconStor StorSafe server via iSCSI to back up the OS boot image and application data on virtual tapes. The IBM i host client needs to be prepared for migration; two virtual tapes are needed; the first tape will contain the base operating system for the initial remote boot and the second tape will contain the remaining data of the host machine image.
2. The primary StorSafe server deduplicates data and replicates both tapes to another StorSafe server in IBM Cloud.
3. On the replica site, the replica virtual tapes are promoted; the first tape is assigned to an IBM i remote boot server via iSCSI.
4. The IBM i remote server in the cloud mounts the OS image on the first promoted tape on an NFS share, then it boots another IBM i virtual server from the NFS boot directory running the same OS as the IBM i of the primary site.
5. The second tape is assigned to the new IBM i via iSCSI. This new server will read data from this second tape to complete the workload migration and become the clone of the IBM i on the primary site.

Refer to <https://www.falconstor.com/partners/ibm-partnership/ibm-i-migration-to-the-cloud-as-a-service>

and the FalconStor knowledge base article **KB 2162**, which contains custom scripts developed for migration of IBM i/AIX workloads from on-premises Power to the PowerVS Cloud.

### Support Matrix for D-Mode IPL (Bootimg)

A D-Mode Initial Program Load (IPL) from an iSCSI StorSafe tape is supported under the following conditions:

| IBM i Level          | Power System Level        | D-Mode IPL Supported |
|----------------------|---------------------------|----------------------|
| 7.5 base and earlier | Any                       | No                   |
| 7.5 TR1 or later     | Power10 (FW1030 or later) | Yes                  |
| 7.6                  | Power10 (FW1060 or later) | Yes                  |

### When D-Mode IPL is Supported

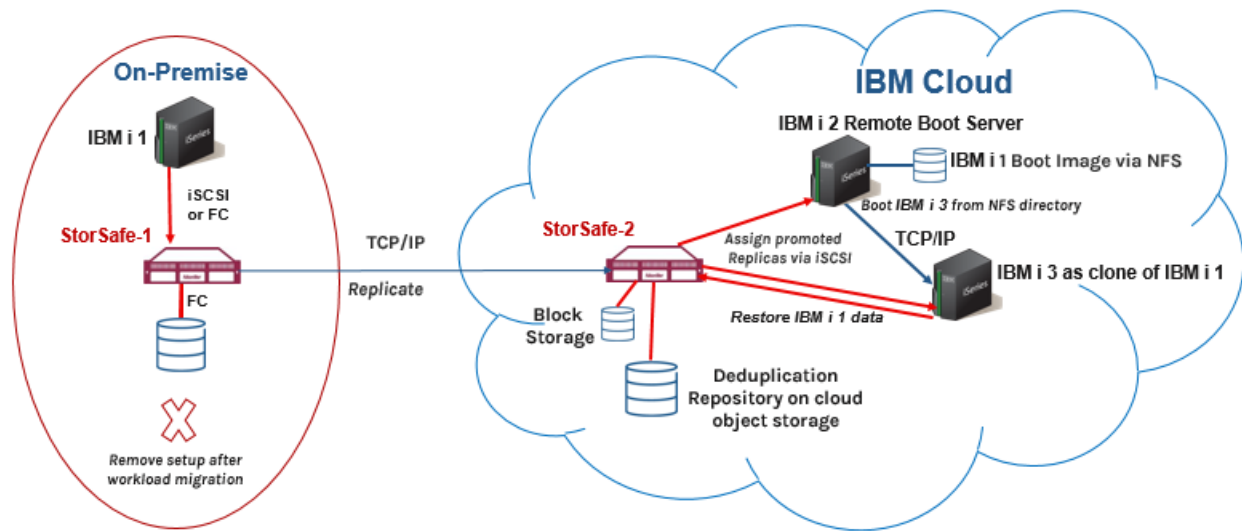
If your environment meets the requirements above, you can perform a full system save/restore using the iSCSI StorSafe tape in a manner similar to a physical tape library. However, there are specific procedural differences:

1. **HMC Network Boot Wizard:** You must use the HMC Network Boot wizard to initiate the IPL because configuration information must be entered before the process begins.
2. **TCP/IP Activation:** Extra steps are required to activate TCP/IP while in a restricted state.
3. **Security Limitation:** During the restoration of the Licensed Internal Code (LIC) and operating system components, only a limited function TCP/IP stack is available, which does not support secure connections. Sensitive user data should only be restored after the full operating system TCP/IP stack is started and secure connections are configured.

### When D-Mode IPL is Not Supported

If the operating system or hardware level does not support D-Mode IPL from the iSCSI StorSafe tape, you must use an alternate source for the initial boot. The process generally involves:

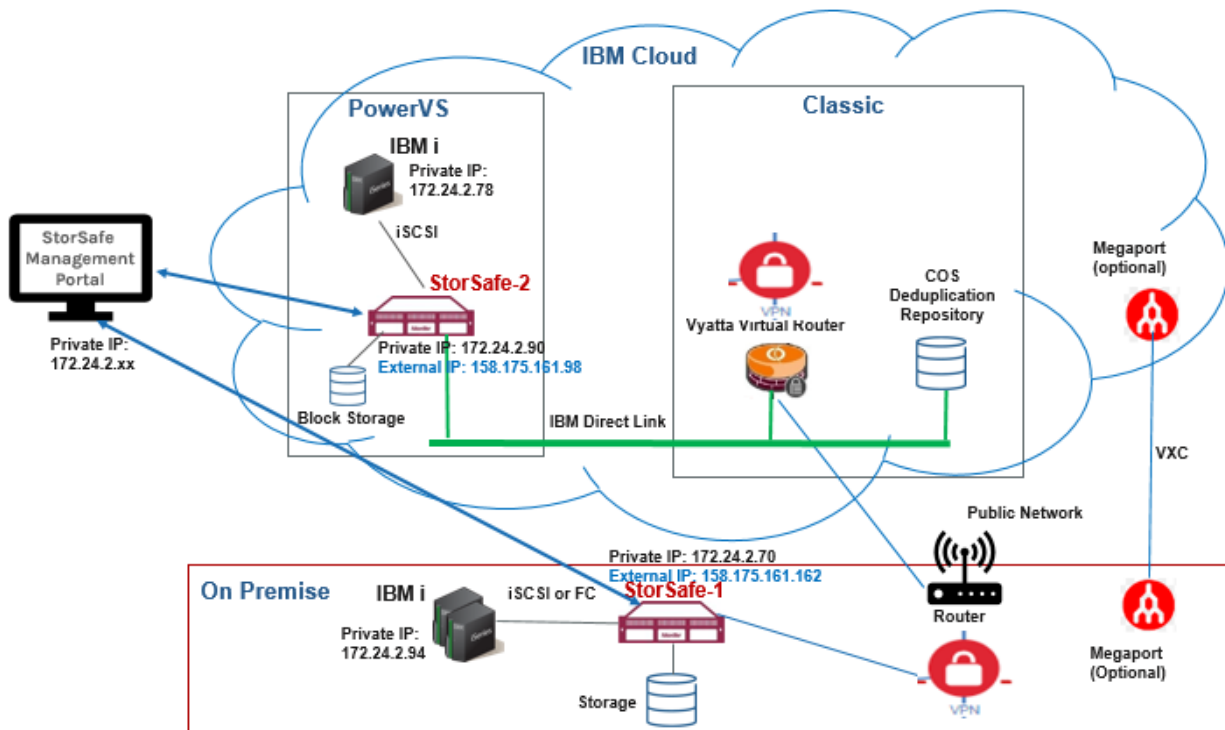
1. Creating a virtual optical image catalog (IMGCLG) containing the SAVSYS and base operating system.
2. Hosting that image on an NFS server (another IBM i partition) to perform an NFS D-mode IPL/install.
3. Once the target system is IPLed and TCP/IP is started, the iSCSI StorSafe tape is then used to restore the remaining bulk data.



## Connectivity

The following connections are required:

- Network connection between the on-premises StorSafe and the PowerVS StorSafe in IBM Cloud
- Network connection between StorSight in the x86 infrastructure and the on-premises StorSafe
- Network connections between StorSight in the x86 infrastructure and the PowerVS StorSafe in IBM Cloud
- iSCSI or Fibre Channel connection between IBM i host clients and the on-premises StorSafe
- iSCSI connections between IBM i host clients and the PowerVS StorSafe in IBM Cloud
- IBM Cloud Direct Link connection or site VPN between the PowerVS StorSafe and the IBM COS
- Internet connection with the FalconStor license server for online registration (optional)
- When an IBM i host client is used for a workload migration scenario, it needs to be an NFS server to use the boot image for the cloned machine. Refer to IBM documentation for more information.



# Prepare Deployment

This section contains information and deployment guidelines for FalconStor StorSafe in IBM Cloud.

## Outline of steps

The following highlights the steps you need to perform; for each step, detailed information is provided in sections below:

1. Obtain an IBM cloud account with adequate access rights.
2. Identify the solution sizing.
3. Create worksheets to plan and record your deployment configuration.
4. Create or use an existing Power Systems Virtual Server service that represents your datacenter.
5. Create at least one subnet in your Power Systems Virtual Server service. You can create two more subnets to isolate traffic via VLANs for backup data and replication data.
6. Create SSH keys in the Power Systems Virtual Server service for secure remote connections.
7. Use the *FalconStor StorSafe for PowerVS* tile in the IBM catalog to create a power virtual server running the FalconStor software.
8. If you use COS for the deduplication data repository, create an object storage bucket on IBM Cloud. Note the access key, password, endpoint specifying the full URI/URL path to the object storage, region, and bucket name.
9. Install the FalconStor StorSight management portal.
10. After you create a StorSafe PowerVS, look for an email from FalconStor to receive a license keycode to enter into FalconStor StorSight. StorSafe servers do not require individual licenses. A global unique license is provided for StorSight that manages all StorSafe servers. The FalconStor StorSafe license will be activated as soon as it is added to StorSight for management.
11. Configure FalconStor StorSafe via the StorSight management portal.
12. Create power servers running backup software that will be attached as iSCSI clients to FalconStor StorSafe.
13. Configure iSCSI clients to be attached to the FalconStor StorSafe iSCSI target to perform backups.

Verify the following components before starting the deployment:

- Network connectivity
- Storage connectivity and integrity
- While you should have the current, generally available version of the software product at the time of deployment, you should always check the FalconStor Portal for the latest revisions and patches.

## Access rights

Verify that you have Manager service access role for IBM Cloud Schematics.

Review and verify the Identity and Access Management (IAM) information to confirm you have adequate rights to create resources.

## Sizing and licensing

To identify the required system resources, such as the amount of memory, CPU cores, and storage capacity for your FalconStor servers, you need to follow the sizing guidelines. StorSafe sizing depends on the amount of data to retain based on the size of ingest data, deduplication ratio, and backup parameters.

Refer to the FalconStor [Solution Sizing](#) page to get the sizing information about the deduplication repository, backup cache, memory, CPU, and the machine type.

Refer to the IBM [Cost Estimation](#) page. Click *Estimate costs* on the top right-side panel, select *Virtual Tape Library* as the operating system, enter values for usage parameters based on the Solution Sizing tool results, and then click *Calculate cost* and *Save* to see the cost. Click *Review estimate* to go to the *Cost estimator* page. Additional costs may apply based on extra capacity for the COS, or additional network and infrastructure components. For the COS capacity, go to *Catalog*, type *Object Storage* in the Search box, and select the *Standard* plan. Click *Estimate costs*, enter a value for *Monthly average capacity*, and click *Calculate cost* again.

## Required storage

Besides the operating system (OS) disk, there are four classes of storage resources:

- Tapes – For storing virtual tapes and virtual index tapes
- Configuration Repository and Tape Database - A small resource used for reporting and configuration management
- Deduplication Repository – For storing unique blocks of data
- Deduplication Index/Folder - For storing deduplication metadata

The following storage types are required for StorSafe. Only the deduplication repository can use cloud object storage; others use block storage:

- OS boot disk: 200 GB
- Configuration repository and tape database disk: 20 GB
- Deduplication repository: Amount of data to retain based on the size of ingest data, deduplication ratio, and backup parameters; the deduplication ratio is an estimated value based on the nature of the data. To estimate the deduplication repository size, you can divide the amount of data by the deduplication ratio.
- Storage for deduplication index and folder disks: 4.3% of the deduplication repository size
- Storage as backup cache to hold virtual tape data: Sum of:
  - a. Amount of data to be moved to the deduplication repository. If the input rate is high, the cache must be large enough to buffer data for several days. To be safe, you can estimate the amount as one week's worth of data, considered as *weekly ingest data*.
  - b. Size of virtual tape indexes, which is estimated as 3% of *weekly ingest data* multiplied by the number of weeks to retain data
  - c. Size of replica tape indexes in case of incoming replication, which is estimated as 6% of *weekly ingest data* multiplied by the number of weeks to retain data
  - d. Size of largest dataset to restore

## Object Storage

If you use Cloud Object Storage (COS) for the StorSafe deduplication repository data disks, get an S3 object storage bucket with "**Writer**" access, with service credentials including HMAC. Note the bucket name, location, access key ID, and secret access key. Create an IBM COS bucket with the service credentials, including HMAC for the deduplication repository. You can use "*Storage class: Standard*" and "*Resiliency: Regional*" settings.

## Required memory

The memory reserved for deduplication depends on the size of the deduplication repository, where 2 GB memory is used for each 1 TB of deduplication repository capacity.

You will need at least 16 GB of base memory plus 2 GB for each TB of deduplication repository.

## Required CPU cores

A core is a physical unit of a Central Processing Unit (CPU) that acts as a separate processing unit. Basically, a core can be considered as a small processor built into the main processor that is connected to a socket. A virtual CPU (vCPU), also known as a virtual processor, represents the central processing unit used in a virtual machine (VM). Each vCPU in a VM operating system represents one physical CPU core. Since there is a core-to-vCPU ratio of 1:1, for shared processors, fractional cores round up to the nearest whole number. For example, 1.25 cores equal 2 vCPUs. Note that with hyper-threading the OS detects a single-core processor as a processor with more logical cores (not physical cores).

For example, the `lscpu` command shows the following output for a VM with one core:

```
[root@ss1113ga ~]# lscpu
Architecture:      ppc64le
Byte Order:        Little Endian
CPU(s):            8
On-line CPU(s) list: 0-7
Thread(s) per core: 8
Core(s) per socket: 1
Socket(s):         1
NUMA node(s):     1
Model:            2.2 (pvr 004e 0202)
Model name:       POWER9 (architected), altivec supported
Hypervisor vendor: pHyp
Virtualization type: para
L1d cache:        32K
L1i cache:        32K
NUMA node0 CPU(s): 0-7
Physical sockets:  2
Physical chips:    1
Physical cores/chip: 10
[root@ss1113ga ~]#
```

For large systems requiring a high amount of memory based on the deduplication repository size, use more CPU cores:

- Deduplication repository up to 10 TB: 1 CPU core
- Deduplication repository up to 50 TB: 2 CPU cores
- Deduplication repository up to 100 TB: 4 CPU cores
- Deduplication repository over 100 TB: 8 CPU cores

## Network connections

The StorSafe instance can have public and private networks for remote access. The public network hosts an external IP address. You can either connect to the instance using the IBM **Open console** option or extend the private network for internet access to StorSafe via a remote SSH session using the pre-installed SSH key. Once you are connected you can configure required routing to private IP interfaces for private routes.

The traffic with the StorSight management portal, IBM COS for deduplication repository, iSCSI clients, and tape replication goes through the private IP addresses. Perform the following steps:

- Configure network routers with a VPN configuration and firewall settings.
- If IBM COS is used for the deduplication repository, set up necessary connections such as VPC and private endpoints, VPN connection, or cloud connection, depending on the network requirements in your use case.

- Check network settings for speed and Maximum Transmission Unit (MTU) values on network devices and routers. For example, network devices can have an MTU value of 1500 bytes but tunnel interfaces, as used in the cloud, can have **1476** or lower bytes, since they use some bytes for IP headers. You can use the system command `ping` to confirm the maximum transmission size value for a network device (default `eth0`) does not display any errors:  

```
# ping [-I <NetWork device>] -s <MTU value> -M do <IP address>
```
- Make sure the DNS settings are correct in `/etc/resolv.conf` of the StorSafe and StorSight servers to resolve the host names. Use the system command `ping` to confirm connectivity with the name servers and FalconStor servers. For private DNS servers, consult the IBM documentation from <https://ondeck.console.cloud.ibm.com/docs/dns?topic=dns-dns-faq>.

## Required network ports

Make sure network firewalls allow StorSafe access through the following ports.

Refer to the *Appendix* section in the *FalconStor StorSafe with StorSight User Guide* for more details:

- TCP port 22 for remote connection via Secure Standard Shell (SSH)
- TCP/UDP port 25 for email alerts via SMTP
- TCP port 80 for internet connection to the FalconStor license server (*register.falconstor.com*), for online registration of license keycodes. If this cannot be set up for security reasons, offline registration can be used.
- UDP port 123 for time synchronization via NTP
- TCP port 3260 for communication between StorSafe and iSCSI clients
- TCP port 11576 for secure RPC communication between StorSafe and StorSight
- TCP/UDP port 11577 for incoming data replication (if replication is configured)
- TCP/UDP port 11579 for replication authentication (if replication is configured)
- TCP port 11581 for StorSight statistics gathering
- TCP port 11582 for CLI commands (if applicable)
- TCP port 11584 for replication of deduplicated data (if replication is configured)
- TCP port 11583 for report requests
- TCP ports 11781 and 11782 for replication encryption (if applicable)
- TCP port 18651 for non-encrypted replication (if applicable)
- TCP port 18652 for encrypted replication (if applicable)

### StorSight ports

- HTTP port 80 for internet connection
- HTTPS port 443 for secure internet connection
- LDAP port 636 for LDAP and Active Directory authentication (if applicable)
- TCP port 11753 for retrieving StorSafe configuration

## Performance Considerations

### Network

You can isolate network traffic using different Virtual Local Area Networks (VLANs). For example, you may want to prevent access from the VLAN that connects to IBM COS from the VLAN used by IBM i host clients.

For better performance, use separate network interfaces to isolate traffic between VLANs; you can also dedicate a different adapter for replication.

You can set up VLANs to isolate network traffic between:

- IBM i host clients and StorSafe for ingest data
- StorSafe and StorSight for management
- StorSafe and IBM COS for deduplicated data
- StorSafe source and replica servers for data replication

If several host systems are connected to different network interfaces on StorSafe, assign separate network segments to each interface to isolate the iSCSI traffic and avoid timeouts and frequent reconnections.

Make sure all network paths maintain similar speeds across all devices and routers between the backup clients and the StorSafe server, as well as between the StorSafe server and the object storage. If all network devices and routers support Jumbo Frames, enable Jumbo Frames and tune the Maximum Transmission Unit (MTU) for optimal performance. The MTU size in a cloud environment can vary depending on the provider and the specific network configuration.

### Block Storage

Storage Tier 3 is more cost-effective than Tier 1, but Tier 1 provides better I/O performance. Ensure all attached storage devices use the same storage type; do not mix storage types.

### Cloud Object Storage

Using COS as the StorSafe deduplication repository may impact restore performance. While COS is more cost-efficient than block storage, block storage offers significantly better performance. If storage cost is not a concern, opt for block storage for faster restores. If you use COS for the deduplication repository, use Smart Tier or Standard (low latency) storage class. In case you enable the NAS feature and perform random IO to NAS shares, do not use COS for the deduplication repository.

Two mechanisms are available for StorSafe PowerVS access to COS; refer to <https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-per>.

- With a Power Edge Router (PER) and a Network Translation (NAT) device, you can access COS directly via the COS private endpoint, but that is not a recommended path as the throughput is not optimal and for large volumes it can potentially become a network Bottleneck:  
PER -> NAT -> PER -> Private Path Router (PPR) -> COS
- With the IBM Cloud Transit Gateway that acts as a central hub for interconnectivity between StorSafe PowerVS and COS VPC networks, you can connect StorSafe to COS via Virtual Private Endpoint (VPE). It is recommended to use this method, which provides better bandwidth than access through routers. Use a *Direct* endpoint for COS access, which does not incur charges for any outgoing or incoming bandwidth even if the traffic is cross regions or across data centers.  
PER -> Transit Gateway (PPR) -> VPE -> COS

With a proxy server, StorSafe PowerVS will submit COS requests to the IP or URL of the proxy. Make sure the DNS settings are correct in `/etc/resolv.conf` of the StorSafe server to resolve the endpoint name. Use the system command `ping` to confirm connectivity with the endpoint. For private DNS servers, consult the IBM documentation from <https://ondeck.console.cloud.ibm.com/docs/dns?topic=dns-dns-faq>.

### ***Backup Application***

For improved restore performance, use multiple drives on the host client for multi-stream backup/restore operations. Multiple streams increase speed; for example, a single drive may restore at 40 MB/s, whereas using eight drives could achieve 200 MB/s.

On IBM i, you can use the CHGTCPA command to configure various TCP/IP settings, including the TCP receive and send buffer sizes. We recommend half the max size (4194304) for better performance.

- TCP Receive Buffer Size – Determines the amount of memory allocated to store incoming data before the application processes it. A larger buffer can help improve performance by reducing the need for frequent acknowledgments.
- TCP Send Buffer Size – Determines the amount of memory allocated for storing outgoing data before it is sent over the network. A larger send buffer allows data to be transmitted more efficiently, reducing latency in high-speed networks.

On NetBackup, you can set the buffer size for data transfer in `.../db/config/SIZE_DATA_BUFFERS`. Common values are:

- 262144 (256 KB)
- 524288 (512 KB)

## Deployment Worksheets

Use the following worksheets to plan and record your deployment configuration.

### Network IP addresses

The table below describes information about IP addresses needed on each StorSafe server:

- **Name** represents a meaningful label to identify the network usage
- **Port** represents a network device, *ethn*
- **Type** can be
  - *Private* (for private subnets and VLANs)
  - *Public* (for outside communication)
- **Usage** identifies what the network will be used for, such as:
  - StorSight management portal and COS access
  - *iSCSI clients* in different areas
  - *Replication* traffic with a remote server
  - *Outside* communication via a public IP address

| Name | Port | IP Address | Type | Usage |
|------|------|------------|------|-------|
|      |      |            |      |       |
|      |      |            |      |       |
|      |      |            |      |       |
|      |      |            |      |       |
|      |      |            |      |       |
|      |      |            |      |       |

### iSCSI clients

The table below describes information about iSCSI clients.

| Client Name | IP Address | iSCSI Initiator IQN | iSCSI Target IQN |
|-------------|------------|---------------------|------------------|
|             |            |                     |                  |
|             |            |                     |                  |
|             |            |                     |                  |
|             |            |                     |                  |

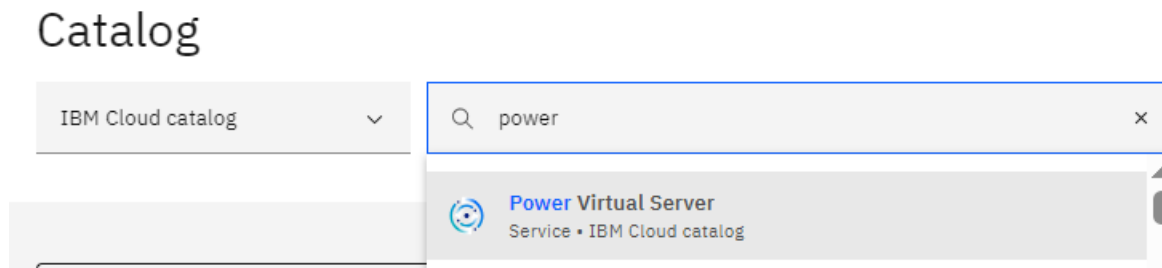
### Data replication

|                          |  |
|--------------------------|--|
| Source Site Location     |  |
| Target Site Location     |  |
| WAN Link Type            |  |
| WAN Length               |  |
| WAN Bandwidth            |  |
| Cross Replication or not |  |
| Dedicated or Shared      |  |

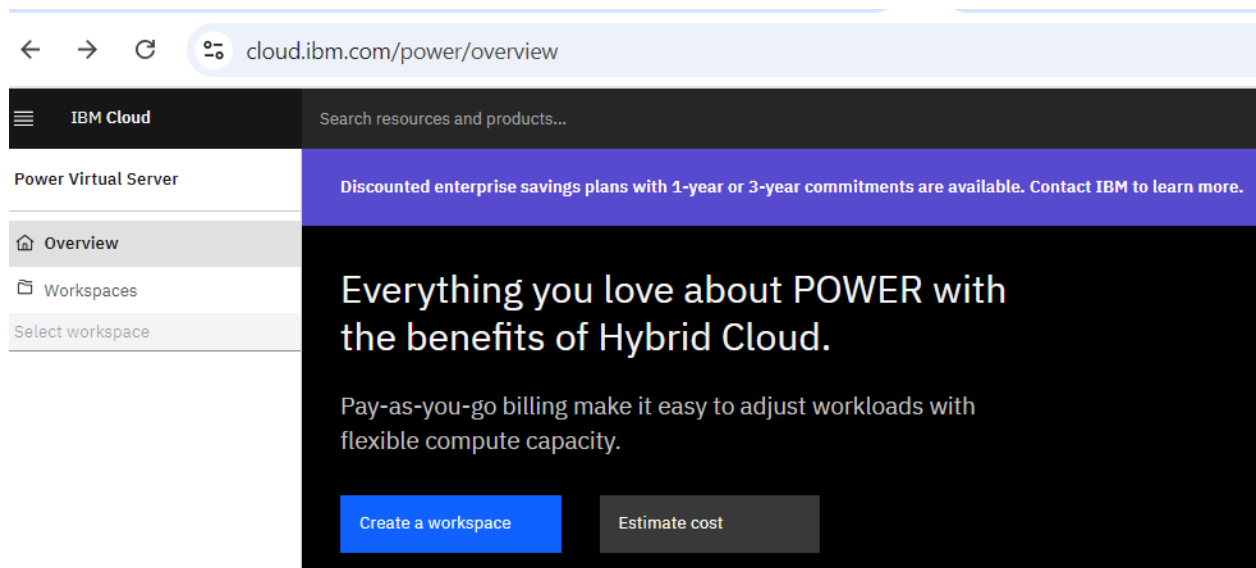
## Create a Power Systems Virtual Server service

Before deploying a StorSafe PowerVS instance, you need to have a Power Systems Virtual Server service that represents a datacenter with adequate networking and infrastructure for your deployment scenario. If you do not have a Power Systems Virtual Server service, follow the instructions below to create one.

1. Connect to IBM Cloud using your credentials.
2. Click *Catalog* in the menu bar, type *Power* in the search box, and select *Power System Virtual Servers*.



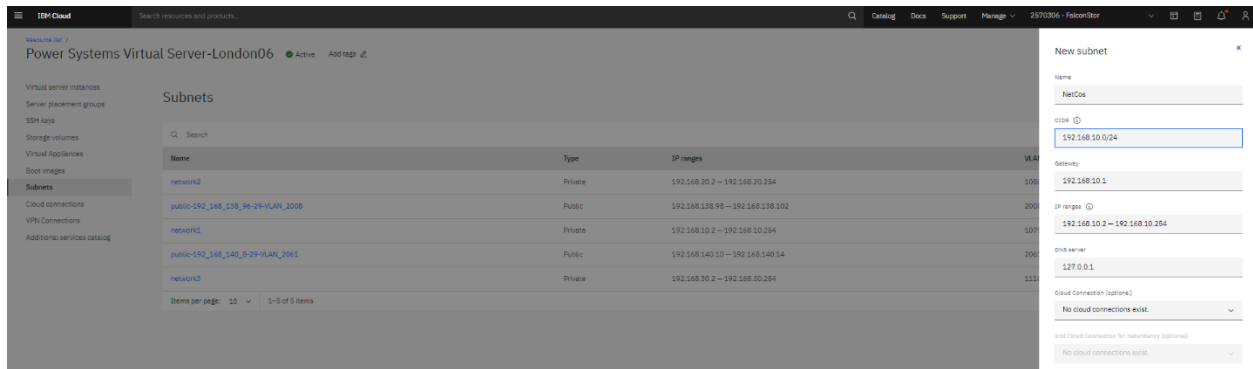
3. Create a workspace in your region.



## Create network subnets

To isolate traffic via VLANs, create at least three subnets in your Power Systems Virtual Server workspace for different usage, such as *StorSight Portal* and *COS* access, *iSCSI clients* in different areas, and *Replication* traffic with a remote server. You can have as many subnets as you require in each Power Systems Virtual Server service. Once you specify the networks, IBM *cloud-init* configures the IP addresses.

1. Click *Resource list*, select *Compute*, and select your Power Systems Virtual Server workspace.
2. Select *Subnets* and click *Create subnet*.
3. Enter a meaningful label to identify the network usage.
4. Enter the IP address range and click *Create a subnet*.



## Add SSH keys

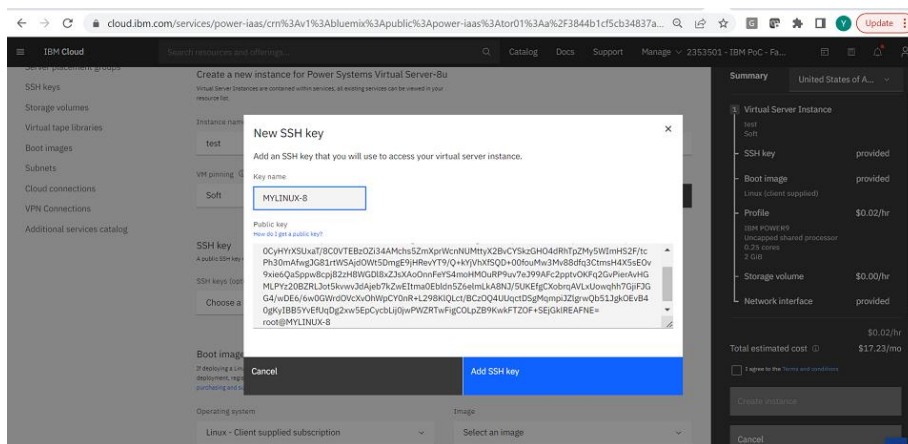
SSH keys are used for secure remote connections to StorSafe.

A public key can be created by using the `ssh-keygen` command available in the *OpenSSH Client* application to generate the RSA public key file `id_rsa.pub`:

```
# ssh-keygen
# cat /root/.ssh/id_rsa.pub
ssh-rsa
AAAAB4NzaC1yc2EAAAADAQABAAQGBGZx4Z8z8AskuIgVOGBJ+psPbmOOHCAApR3bfoOHDztyG36rt
0CyHYrXSUxaT/8C0VTEBzOZi34AMchs5ZmXprWcnNUMttyX2BvCYSkzGH04dRhTpZMy5WImHS2F/t
cPh30mAfWgJG81rtWSAjdOWt5DmgE9jHRevYT9/Q+kYjVhXfSQD+00fouMw3Mv88dfq3CtmsH4X5s
EOv9xie6QaSpw8cpj82zH8WGD18xZJsXAOonnFeYS4moHMOuRP9uv7eJ99AFc2pptvOKFq2GvPie
rAvHGMLPYz20BZRLJot5kvwvJdAjeB7kZwEItma0EblDn5Z6elmLkA8NJ/5UKEfgCXobrqAVLxUow
qhh7GjiFJGG4/wDE6/6w0GWrdOVcXvOhWpCY0nR+L298KlQLct/BCzOQ4UUqctDSgMqmpiJZlgrwQ
b51JgkOEVB40gKyIBB5YvEfUqDg2xw5EpCycbLij0jwPWZRTwFigCOLpZB9KwkFTZOF+SEjGklREA
FNE= root@MYLINUX-8
```

To add your SSH key to the Power Systems Virtual Server service, perform the following steps:

1. Click **Resource list**, select **Compute**, and select your Power Systems Virtual Server workspace.
2. Select **SSH keys** and click **Add SSH key**.
3. Enter a **Key name** and copy and paste your SSH key file contents in the **Public key** area. Click **Add SSH key**.



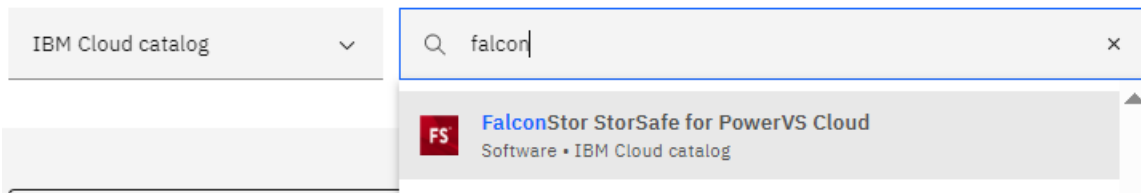
# Create StorSafe Power Virtual Server

This section explains how to create a power virtual server running the FalconStor StorSafe software product in IBM Cloud. The FalconStor Open Virtual Appliance (OVA) package contains the Linux OS image and FalconStor StorSafe software.

Follow the instructions below to access the software from the IBM catalog:

1. Connect to IBM Cloud using your credentials.
2. Click *Catalog* in the menu bar, type *falcon* in the search box, and select the *FalconStor StorSafe for PowerVS* tile.

## Catalog



3. Leave default settings for the deployment target, the delivery method, the product version. You can update the default workspace name. Leave the default settings for the location or select another location in the list that is close to your region where the terraform schematics will be executed. Leave other settings as default.

A screenshot of the FalconStor StorSafe for PowerVS Cloud deployment configuration page. The page is divided into several sections. On the left, there is a sidebar with links to 'Catalog', 'Compute', 'Network', 'Storage', and 'Security'. The main content area is titled 'FalconStor StorSafe for PowerVS Cloud' and contains the following sections: 'Select a delivery method' (with 'Power Systems Virtual Server' selected), 'Select product version' (with '11.13.2' selected), 'Review your pricing plans' (with a table showing pricing plans), 'Configure your workspace' (with fields for 'Name', 'Location', 'Resource group', and 'Type'), and 'Summary' (with a table showing deployment details). The 'Name' field is filled with 'vltile-11.13.2-01-01-2023'. The 'Location' dropdown is set to 'London (eu-gb)'. The 'Resource group' dropdown is set to 'Default'. The 'Type' dropdown is set to 'Example: 11.13.2-01-01-2023'. The 'Summary' section on the right contains a table with the following information: 'Deployment target: Power Systems Virtual Server', 'Delivery method: Server Image', 'Workspace: vltile-11.13.2-01-01-2023', 'Resource group: Default', and 'Location: London (eu-gb)'. There is also a note about 'Information required' and a link to 'View the existing installations'.

4. Set the server required values:
  - a. Select the Power Systems Virtual Server Cloud Resource name (CRN) that represents your data center.
  - b. Enter a name for your StorSafe instance.
  - c. Set the memory size, according to the sizing calculator tool results.
  - d. Enter one network name that you want to use, as defined for the selected Power Systems Virtual Server CRN. Refer to the [Network IP addresses](#) worksheet that has been prepared for this deployment.

- e. Set the size of your license repository capacity. Make sure that the license repository capacity is set to a value higher than zero and is aligned with the repository size.
- f. Enter the SSH key name, as defined for the selected Power Systems Virtual Server CRN.
- g. Choose the storage type; *Tier 3* is cheaper than *Tier 1* but *Tier 1* provides better I/O performance. Make sure you choose the same storage type for all attached storage devices; do not use mix storage types. However, change for the Configuration Repository storage type to “Fixed IOPS” a higher number of I/O operations and performance improvement.
- h. Select the type of the system on which to create the StorSafe VM based on the server sizing: *s922* or *e980* for Power 9 and *s1022* or *e1050* for Power 10. For *s1022/e1050* make sure a Power10 machine exists on the datacenter that you selected in CRN. The system type *e980/e1050* should be used for a large system that requires more memory to hold a deduplication repository of around 400 TB.
- i. Set the number of vCPU cores, according to the sizing calculator tool results.

## Required input variables

A value for each of the following parameters is required. A default value might be set for some parameters. You can choose to accept the default value or update it.

| Parameter           | Description                                                                                                                                                            | Value               |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| crn                 | Power Systems Virtual Server CRN                                                                                                                                       | Select a value ▼    |
| instance_name       | The name to assign to the StorSafe instance                                                                                                                            | Enter instance_name |
| memory              | The amount of memory to assign to the StorSafe instance in GB according to the following formula: $\text{memory} \geq 16 + (2 * \text{license\_repository\_capacity})$ | 18 -   +            |
| network_1           | The first network ID or name to assign to the StorSafe instance, as defined for the selected Power Systems Virtual Server CRN                                          | Enter network_1     |
| repository_capacity | The StorSafe licensed repository capacity in TB; must be higher than zero                                                                                              | 1 -   +             |
| ssh_key_name        | The name of the public SSH RSA key to access the StorSafe instance, as defined for the selected Power Systems Virtual Server CRN                                       | Enter ssh_key_name  |
| storage_type        | The type of storage tier for all volumes to attach to the StorSafe instance: 'tier1' (high performance) or 'tier3'                                                     | tier1               |
| system_type         | The type of system on which to create the StorSafe instance: 's922' or 'e980' for Power 9; 's1022' or 'e1050' for Power 10 if present in the selected datacenter       | s922                |
| vcpus               | The number of vCPUs, AKA virtual processors, to assign to the StorSafe instance; one vCPU is equal to one physical CPU core.                                           | 1 -   +             |

5. Expand the *Optional input values* to check other deployment values.
  - a. You can enter two extra network names that you want to use for traffic isolation.
  - b. You can enter a specific IP address for each network. This is useful when you perform an A-to-B upgrade scenario, where you want the network configuration of source machine A be preserved on target machine B.
  - c. If applicable, enter the name of a server placement group where the StorSafe instance should be placed, as defined for the selected Power Systems Virtual Server CRN.
  - d. Set the processor mode: *Shared*, *Dedicated*, or *Shared Capped*. *Shared* is less expensive.
  - e. To place the StorSafe volume on storage controllers, according to a storage anti-affinity policy, enter the PVM ID of other server instances, as defined on the selected Power Systems Virtual Server CRN. The ID is displayed in the server details. This can be useful when other VMs exist in your environment and StorSafe is not the first VM to be added. An anti-affinity rule places a group of virtual machines across different storage controllers, which prevents all VMs from failing at once in case a single controller fails.

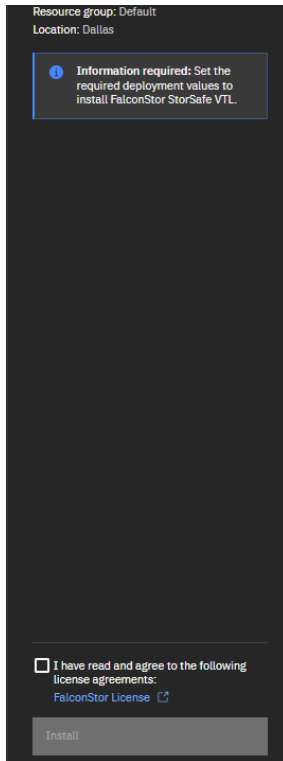
NOTE: For both storage affinity and server placement group, we recommend that StorSafe not be on the same server or storage box as other machines it will be backing up.

- f. Accept or update block storage size values for the configuration repository, tape cache, deduplication repository index and folder disks.

**Optional input variables**  
You can enter a value for the following parameters. Some might include default values, which you can accept or update.

| Parameter                 | Description                                                                                                                                                                                        | Value                                              |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| network_1_ip              | Specific IP address to assign to the first network rather than automatic assignment within the IP range                                                                                            | <input type="text" value="Enter network_1_ip"/>    |
| network_2                 | The second network ID or name to assign to the StorSafe VTL instance, as defined for the selected Power Systems Virtual Server CRN                                                                 | <input type="text" value="Enter network_2"/>       |
| network_2_ip              | Specific IP address to assign to the second network rather than automatic assignment within the IP range                                                                                           | <input type="text" value="Enter network_2_ip"/>    |
| network_3                 | The third network ID or name to assign to the StorSafe VTL instance, as defined for the selected Power Systems Virtual Server CRN                                                                  | <input type="text" value="Enter network_3"/>       |
| network_3_ip              | Specific IP address to assign to the third network rather than automatic assignment within the IP range                                                                                            | <input type="text" value="Enter network_3_ip"/>    |
| placement_group           | The server placement group name where the StorSafe VTL instance will be placed, as defined for the selected Power Systems Virtual Server CRN                                                       | <input type="text" value="Enter placement_group"/> |
| processor_mode            | The type of processor mode in which the StorSafe VTL instance will run: 'shared', 'capped', or 'dedicated'                                                                                         | <input type="text" value="shared"/>                |
| pvm_instances             | The comma-separated list of PVM instance IDs for the storage anti-affinity policy used for placement of the StorSafe instance volume, as defined for the selected Power Systems Virtual Server CRN | <input type="text" value="Enter pvm_instances"/>   |
| volume_configuration_size | The size of the block storage volume for the StorSafe VTL Configuration Repository in GB                                                                                                           | <input type="text" value="20"/> -   +              |
| volume_index_size         | The size of the block storage volume for the index of StorSafe VTL Deduplication Repository in GB; the maximum size of a volume is 2 TB; attach extra volumes later, if necessary                  | <input type="text" value="1024"/> -   +            |
| volume_tape_size          | The size of the block storage volume for the StorSafe VTL tape backup cache in GB; the maximum size of a volume is 2 TB; attach extra volumes later, if necessary                                  | <input type="text" value="1024"/> -   +            |

6. Check all parameters, click the license agreement box on the right, and click *Install* to continue. Once installation is complete, the StorSafe instance will appear in the list of *Virtual Appliances*.



7. Click *Virtual Appliances* and select your instance.
8. Click *StorSafe actions* and select *Open console*. Connect to the instance using the *root* user account and the default password *IPStor101*. The system will then prompt you to enter a new password for the root account, since the default password expires after installation. You can also connect to the IP address via *ssh* with the SSH key using the *centos* user account (*ssh centos@<StorSafe IP Address>*), become *root* by the *sudo su -* command and then run the *passwd* command to enter a new password for the *root* account, to replace the expired password. Alternatively, you can use the *sudo -i* command to enter a new password.
9. Check the network IP addresses with the *ifconfig* command, check the default gateway with the *ip -route* command, and the routing table with the *netstat -nr* command. The default gateway is set based on the first assigned IP subnet that is used for management and can be reached from outside. When there are multiple subnets, you can set the system to use a specific subnet for backup and another subnet for replication.
10. When you deploy a VM on Power, an IPV6 private network connection gets configured in order to manage the VM. If that connection is down, the server may remain in the 'Warning' state and adding disks or changing memory/vCPU cannot be performed. If this occurs, run the following commands to reset the configuration:
 

```
/usr/sbin/rsct/install/bin/uncfgct -n
sleep 5
/usr/sbin/rsct/install/bin/cfgct
sleep 5
/usr/sbin/rsct/bin/rmcctrl -z
/usr/sbin/rsct/bin/rmcctrl -A
/usr/sbin/rsct/bin/rmcctrl -p
```
11. Once the server VM is in an 'Active' state, select the server and click *Attach existing* to add storage volumes to the server, if they are not already attached.

# Deploy StorSafe On-Premises

If you need to deploy an on-premises StorSafe server, select the *FalconStor StorSafe for Power On-Premises* service from the IBM catalog. It can take about 15 minutes to complete.

**FalconStor StorSafe VTL for Power On-Premises**  
Decrease your on-premise backup storage capacity by up to 90%, while improving your existing backup solution performance.

**Create** | About

**Disclaimer**  
This third-party product is provided by a vendor outside of IBM and is subject to a separate agreement between you and the third-party. If you accept their terms, IBM is not responsible for the product and makes no privacy, security, performance, support, or other commitments regarding the product.

**Select a pricing plan**  
Displayed prices do not include tax. Monthly prices shown are for country or location: United States

| Plan       | Features and capabilities                                                                                | Pricing                  |
|------------|----------------------------------------------------------------------------------------------------------|--------------------------|
| Enterprise | Virtual Tape Library Emulation: Emulate tape libraries and back up data to disk or cloud object storage. | \$60.00 USD/TB per month |

**Configure your resource**

Service name: FalconStor StorSafe VTL for Power On-Premises-01

Tags: Examples: env:dev, version-1

Capacity: Indicate the repository capacity or the expansion size in TB. 10

Email Address for License Delivery: user@company.com

Summary:  
FalconStor StorSafe VTL for Power On-Premises  
Location: Global  
Plan: Enterprise  
Service name: FalconStor StorSafe VTL for Power On-Premises-01  
Resource group: Default

☐ I have read and agree to the following third-party terms: [Terms](#)

[Create](#) [Add to estimate](#)

After you create the StorSafe software service, a new resource appears in the IBM *Resource list* under the *Migration* category:

| Name                                 | Group   | Location | Product                           | Status | Tags |
|--------------------------------------|---------|----------|-----------------------------------|--------|------|
| FalconStor StorSafe VTL for Power-on | Default | Global   | FalconStor StorSafe VTL for Power | Active |      |

Follow the steps below to deploy an on-premises server:

1. Click the StorSafe service to display download links for FalconStor software and documentation.
2. Download the Software Appliance Kit (SAK) installation packages for FalconStor StorSafe and FalconStor StorSight.
3. Refer to [Sizing and licensing](#) and the StorSafe and StorSight release notes for sizing and requirements.
4. Refer to the FalconStor Server SAK ISO Image Installation Guide to install the FalconStor software.
5. Look for an email from FalconStor to receive a license keycode to enter in the FalconStor StorSight.

# Install StorSight Management Portal

FalconStor StorSight is a web-based portal for centralized management and monitoring of multiple backup and deduplication servers.

For StorSight installations on a virtual appliance in VPC or Classic infrastructure, refer to instructions in the following section.

For StorSight installations on-premises or on a virtual appliance where a supported Linux operating system has already been installed, use the FalconStor Software Appliance Kit (SAK) image, which installs StorSight and configures the Linux operating system for use with StorSight. Refer to *FalconStor SAK Image Installation Guide* for details.

For StorSight installations on a physical appliance where no operating system exists, use the FalconStor USB bootable image, which contains the operating system, chassis management utilities, and StorSight software. Refer to the *FalconStor Physical Appliance Installation Guide* for details.

## Download software

Download the VHD image to your local machine and unzip it.

## Create the object storage for VHD

1. Connect to the IBM Cloud using your credentials.
2. Click *Catalogue* in the menu bar and select *Object Storage* to create an object storage bucket to hold the FalconStor package:
3. Enter a name for the object storage and complete other fields.

Create, manage, and access your storage in globally available data center locations. Public cloud environments are designed for high durability, resiliency and security. IBM-managed infrastructure automatically scales to businesses needs.

Utilize infrastructure from on-premise data centers, at other cloud providers, or in edge networks as a satellite location to IBM Cloud. Deploy, manage and control app workloads with IBM Cloud Object Storage on client-managed infrastructure.

Select a pricing plan  
Displayed prices do not include tax. Monthly prices shown are for country or region: United States

| Plan | Features                                                                                                                                                                                                                                                                                                                                  | Pricing |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Lite | 6 GB Service Instance<br>Storage up to 10 GB free<br>Up to 2,000 objects a month (PDF, CSV, PPT, and 1,000 requests per month)<br>Up to 10,000 objects a month and 10,000 requests per month<br>Up to 10 GB month of Data Retrieval<br>Up to 100 TB aggregate object count<br>Archive to aggregate 1000 across all storage bucket classes | Free    |

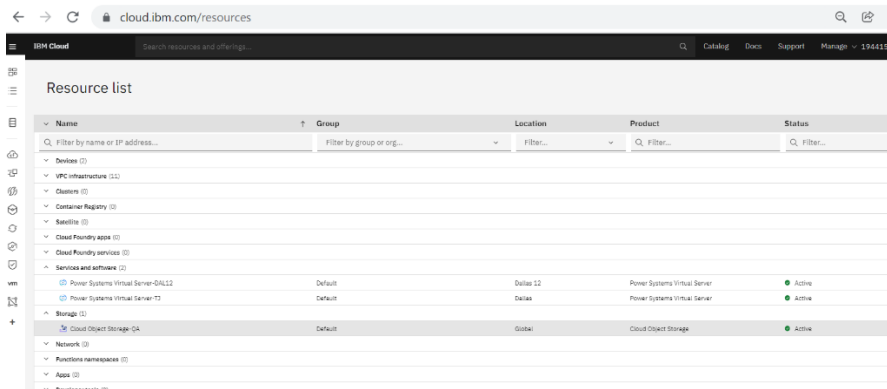
The Lite service plan for Cloud Object Storage includes Regional and cross-Regional resiliency, flexible data classes, and built-in security. The plan resources are deleted after 30 days of inactivity.

Standard There is no minimum fee, so you pay only for what you use. [View Object Storage pricing](#)

Configure your resource

Service name  
Cloud Object Storage-mi

4. Select the object storage that was created.



5. Click *Create bucket*.
6. Select *Customize your bucket*.
7. Enter a name for the bucket and select your location.

**Custom bucket**

Unique bucket name

cos-tj

**Bucket naming rules:**

- Must be unique across the **whole** IBM Cloud Object Storage system
- Do not use any personal information (any part of a name, address, financial or security accounts or SSN)
- Must start and end in alphanumeric characters (3 to 63)
- Characters allowed: lowercase, numbers and nonconsecutive dots and hyphens

**Resiliency**

☐ Cross Region  
Highest availability

☒ Regional  
Best performance

☐ Single Site  
Data sovereignty

**Location** [View options](#)

us-east

**Storage class** [View pricing](#)

**Smart Tier** New!

Smart Tier automatically gives you the lowest storage rate based on your monthly activity.

**Standard**

For active workloads that require higher performance and low latency and where data needs to be accessed frequently.

**Vault**

For less active workloads that require

**Cold Vault**

For cold workloads where data is primarily

8. Select the created bucket and click *Upload* to copy the FalconStor VHD package from your local machine to the bucket.
9. Since the FalconStor VHD package is large, you will need to install the Aspera utility on your local machine. Click *Install Aspera connect*. Then click *Upload files* to open a browser for you to locate the Falconstor VHD package on your local machine for upload.
10. Check the uploaded image in objects list.
11. Click *Service credentials* to create a credential allowing copy of the image from object storage to the boot disk of a power virtual server.

12. Click the down arrow next to the service credentials view the `access_key_id` and `secret_access_key` that you will need to enter for copying the image to your virtual server.

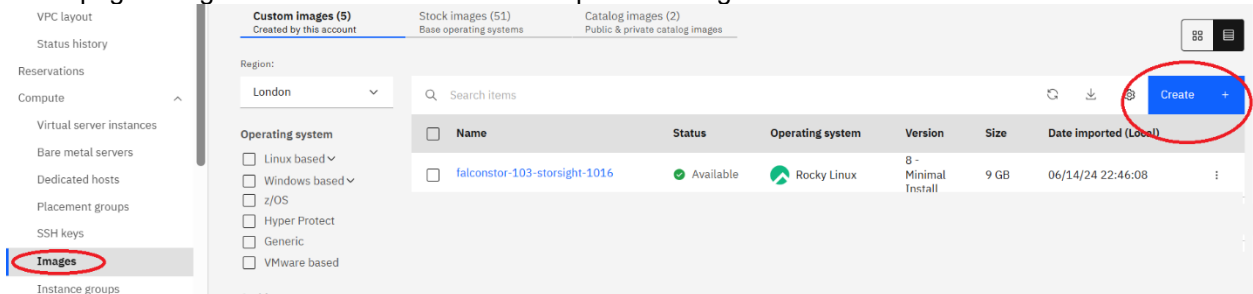
```
{
  "apikey": "qB68Lu0Ub-5ICUnKsk2UwuvrEjMx8vxYE2jq0ehQP6Sz",
  "cos_hmac_keys": {
    "access_key_id": "310-1-144-1-0501fd743",
    "secret_access_key": "e37d97fe3f5f-1-3a32501-0756603d-1-00e827d120b"
  },
  "endpoints": "https://control.cloud-object-storage.cloud.ibm.com/v2/endpoints",
  "iam_apikey_description": "Auto-generated for key crn:v1:bluemix:public:cloud-object-storage:global:a/3844b1cf5cb34837a8c414c73bb08776:67e75043-2bb0-4849-98a6-8a365578aba7:resource-key:318a3bc7-1ad4-4417-9592-9c0521fd743e",
  "iam_apikey_name": "tj-cos",
  "iam_role_crn": "crn:v1:bluemix:public:iam:::serviceRole:Manager",
  "iam_serviceid_crn": "crn:v1:bluemix:public:iam-identity::a/3844b1cf5cb34837a8c414c73bb08776::serviceid:ServiceId-51a995cb-f561-4ecd-b871-147a3214f15c",
  "resource_instance_id": "crn:v1:bluemix:public:cloud-object-storage:global:a/3844b1cf5cb34837a8c414c73bb08776:67e75043-2bb0-4849-98a6-8a365578aba7::"
}
```

**Note:** The upload can take up to an hour. Do not rename the image file. Renaming the file will cause the import to fail.

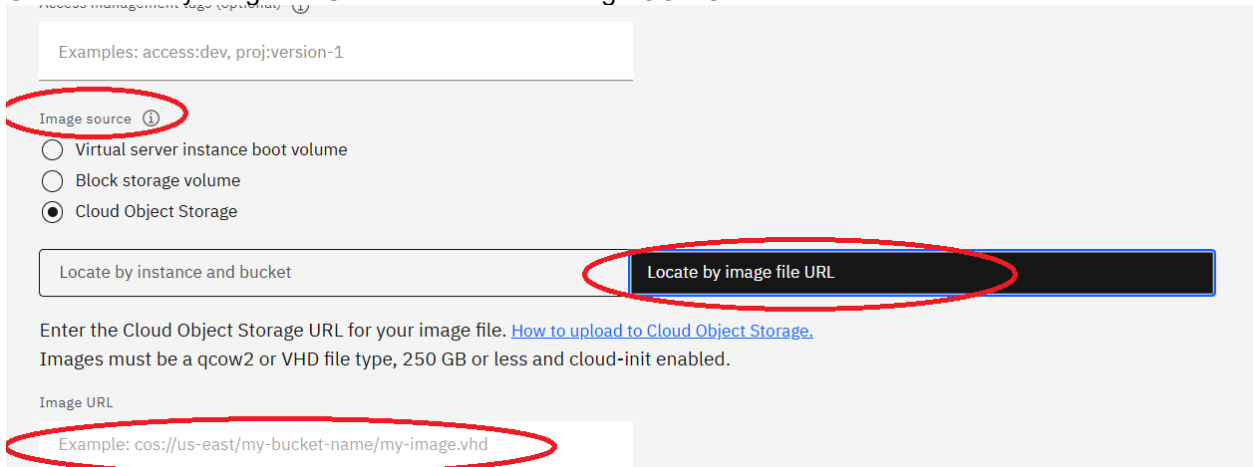
## Install VHD image in VPC

### Import VHD

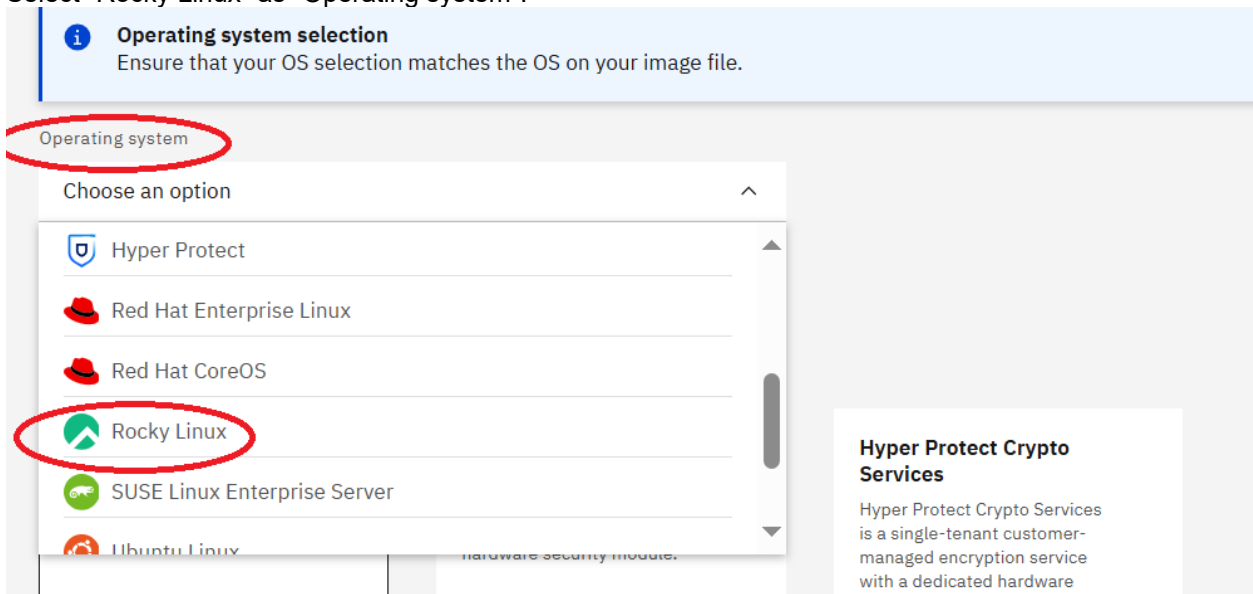
1. Connect to IBM VPC images <https://cloud.ibm.com/vpc-ext/compute/images>.
2. In the page “Images for VPC” select create to import the image.



3. Enter a name for the image you are importing. Select “Cloud Object Storage” as “Image source”. Click “Locate by image file URL” and enter the image COS URL.



4. Select “Rocky Linux” as “Operating system”.



- Click “Create custom image”.

### Create the instance

- On the created image click the three-dots menu and select “Create virtual server instance”.

| Name                                          | Status    | Operating system | Version               | Size  | Date imported (Local) |
|-----------------------------------------------|-----------|------------------|-----------------------|-------|-----------------------|
| <a href="#">falconstor-103-storsight-1016</a> | Available | Rocky Linux      | 8 - Minimal Install   | 9 GB  | 06/14/24 22:46:08     |
| <a href="#">falconstor-102-storsafe-1114</a>  | Available | Rocky Linux      | 8 - Minimal Install   | 15 GB | 06/14/24              |
| <a href="#">falconstor-101-fss902</a>         | Available | CentOS           | 7.x - Minimal Install | 14 GB | 06/11/24              |

- In the “Virtual server for VPC” page enter the virtual server name and click “Change profile” to select the number of required vCPUs and RAM, for example, select *bx2-4x16* for 4 vCPU and 16 GB RAM or *mx2-4x32* for 4 vCPU and 32 GB RAM. Click *Save profile*.

Profile ⓘ

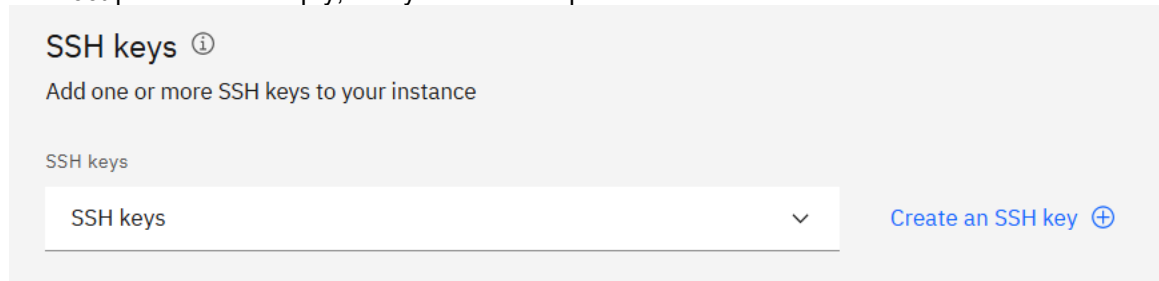
Balanced | bx2-2x8 SAP NetWeaver

[Change profile](#)

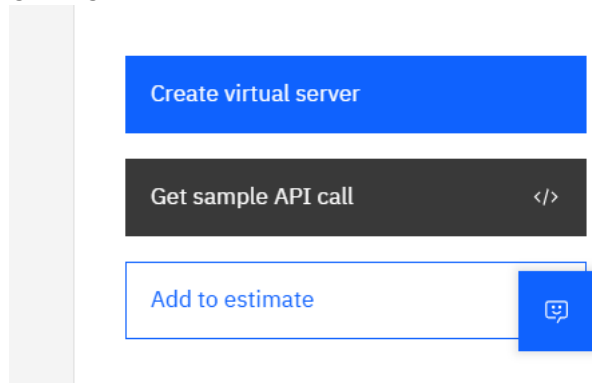
| vCPUs | RAM       | Volume bandwidth | Network bandwidth |
|-------|-----------|------------------|-------------------|
| 2     | 8 GiB RAM | 1 Gbps           | 3 Gbps            |

- Select an existing SSH key click “Create an SSH key” to create a key for this virtual server. The newly created key gets downloaded to your local machine.  
You can connect to the server with the *root* account using this SSH key with the following command:  
`ssh -I <ssh key name> root @<Server IP>`

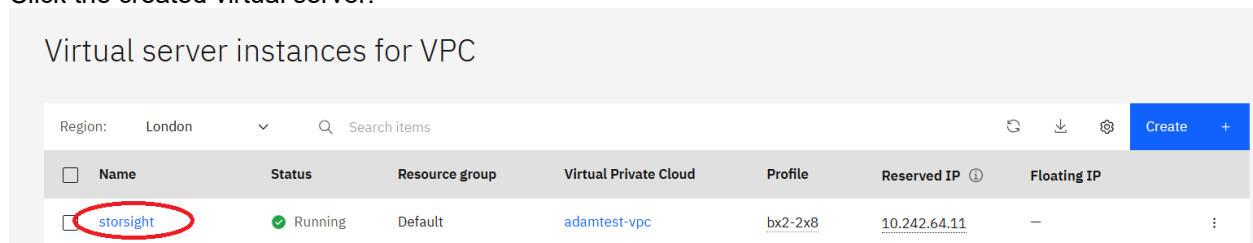
The *root* password is empty, and you can set a password after the first connection.



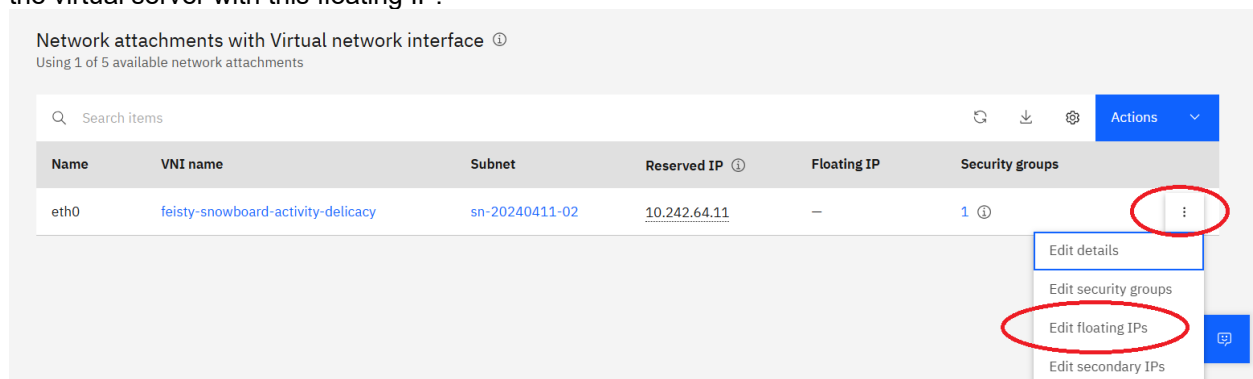
4. Click “Create virtual server”.



5. Click the created virtual server.



6. Click the three-dots menu on the attached network and select “Edit floating IPs”. You can connect to the virtual server with this floating IP.



7. If there is a floating IP, click “Attach”, otherwise click Reserve to get a floating IP.

Edit floating IPs for eth0

×

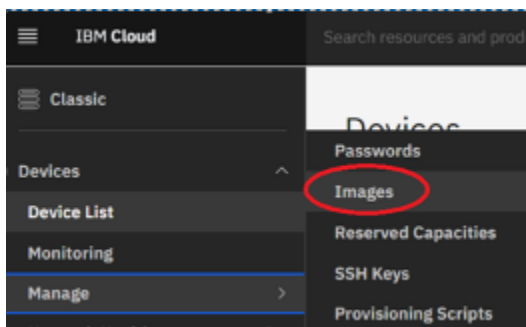
You can add floating IPs to or remove them from the virtual network interface [feisty-snowboard-activity-delicacy](#) that is associated with eth0.

|      |            |  |  |  | <div>Attach +</div> |
|------|------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------|
| Name | IP address |                                                                                     |                                                                                     |                                                                                     |                     |
|      |            |                                                                                     |                                                                                     |                                                                                     |                     |

## Install VHD image in Classic infrastructure

### Import VHD

1. In IBM Cloud, click *Navigation Menu* (☰) > *Infrastructure* > *Classic Infrastructure* > *Manage* > *Images*.



2. Click *Import custom image*.
3. Enter a name for the image and the IBM Cloud API Key for your storage bucket.
4. Select the Cloud Object Storage service instance, the location that is used by your storage bucket, your storage bucket, and the `.vhd` file that you uploaded.
5. From the *Operating system* menu, select *Rocky Linux 8.0 minimal (64-bit)*.
6. From the *Boot mode* menu, select *Hardware Virtual Machine (HVM)*.
7. Check *Your license* and *Cloud-init* boxes.
8. Click *Import image*.

### Create the instance

Follow the steps below to configure network settings and create the server instance.

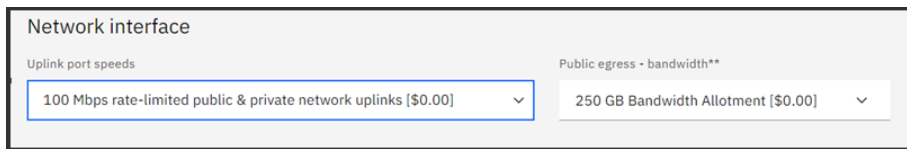
1. Click *Navigation Menu* > *Classic Infrastructure* > *Manage* > *Images*.
2. In the Visibility menu, select *Private images* and find the image that you uploaded.
3. Click *Actions menu* (⋮) > *Order Public VSI*.
4. Select the *Public Multi-tenant* virtual server type.
5. Enter a hostname and domain. The total length of the hostname and domain cannot exceed 64 characters.
6. Select a data center location.
7. Select a profile that meets the data gateway system requirements. Refer to [System requirements for data gateways](#) for more information.
8. Select *View all profile*, select at least 16 GB RAM, and save.

| Profile          |          |              |         |
|------------------|----------|--------------|---------|
| Memory   M1.2x16 |          |              |         |
| 2                | 16 GB    | SAN          | \$84.00 |
| vCPU             | RAM (GB) | Storage type | Price   |

9. Select an SSH key if you have one. Otherwise, select *None*.

10. Choose an uplink port speed under *Public & Private* network uplinks.

You must choose a private-only network. The data gateway will only have a private IP address, and can only be accessed within the same network or through a routing solution of your own choosing.



The screenshot shows a 'Network interface' configuration box. It contains two dropdown menus. The first dropdown, labeled 'Uplink port speeds', is highlighted with a blue border and shows the option '100 Mbps rate-limited public & private network uplinks [\$0.00]'. The second dropdown, labeled 'Public egress - bandwidth\*\*', shows the option '250 GB Bandwidth Allotment [\$0.00]'. Both dropdowns have a small downward arrow icon on the right side.

11. Accept the third-party service agreements and click Create. The Devices screen loads. In a few minutes, a date appears in the Start Date field to confirm that server has started.

# Configure StorSafe via StorSight

You need to run FalconStor StorSight to configure StorSafe. The StorSight web portal allows you to connect to each server for management and monitoring purposes.

Refer to the *FalconStor StorSafe with StorSight User Guide* for more details.

The following highlights the configuration steps for StorSafe:

- Refer to [Required network ports](#) to confirm that required network ports are open for communication between FalconStor software components.
- Open any HTML5-capable browser and type the IP address of your FalconStor StorSight Server. Enter your username, domain, and password to log in. The default login is:
  - Username: *superadmin*
  - Domain: <blank>
  - Password: *freestor*
- Agree to the End User License Agreement (EULA) that appears the first time that the StorSight management portal is launched.
- Select *Administration* -> *StorSight*. Click *License*, and the “+” icon to add your license keycode received from FalconStor. Register your license.
- As per the EULA agreement, it is mandatory to enable the alert notification feature on the StorSight server.  
Select *Administration* -> *Alert Notifications*. Click *Email/SMS Settings*, click the *Email* button, and provide information about your email SMTP server. The *Sender Email* must be a fully qualified email address. Make sure email address *freestorsales@falconstor.com* is whitelisted on your email server. Click *Send Test Email* to verify your configuration.
- Select *Administration* -> *Servers* from the menu bar and click the “+” icon to add your StorSafe server. Enter the IP address or hostname of the StorSafe server. Enter the root user and password for the StorSafe server. Click *Add*.  
When NAT is enabled, both the private IP address and public IP address in the NAT must be present on the StorSafe server being added. An alias network device should be created for the NAT IP address in the system network files. You may need to update the */etc/hosts* file on the StorSafe server to insert a line with the IP address that you used to add StorSafe to StorSight.
- In the *Manage* tab select your StorSafe server.
- Select the *Physical Resources* -> *Physical Devices* tab to virtualize storage devices. If expected devices do not appear, click *Rescan*. Select unassigned physical devices, click the *Manage* icon, and select *Prepare*. Set the category as *Virtual* and the appropriate reservation type: *Configuration* for the 20 GB device to be used as the Configuration Repository and tape database, *Deduplication* for index and folder devices of the Deduplication Repository, or *Tapes* for backup cache devices. Click *Prepare*.
- Select the *Storage Pools* tab. Click the “+” icon to create storage pools. A storage pool is a group of one or more virtualized physical devices (physical devices that have been prepared for use in creating logical resources). Creating a storage pool enables you to provide all of the space needed by your clients in a very efficient manner. Specify a storage pool name and resource types. Select the physical devices to include. Click *Create* to save the storage pool.
- Select the *Object Storage* tab and click the “+” icon to add the object storage account to be used for deduplication. Select the *Generic S3* provider, enter the IBM Cloud access account information noted above, and enable the “*Use for Deduplication Repository*” option. If you use a

reverse proxy server to make remote connections to IBM COS private endpoints, enter the proxy IP address and user/password.

- Select the *Settings* tab.
  - Click the *Configuration Repository* icon. Click *Create Configuration Repository*. Select the physical device reserved for Configuration and click *Create* to enable the Configuration Repository using 10 GB of the reserved device.
  - Click the *Tape Database* icon. Click *Create Tape Database*. Select the physical device reserved for the configuration and click *Create* to enable the tape database using 10 GB of the reserved device.
  - Click the *Deduplication Repository* icon. Click *Create Deduplication Repository*. Select *Object Storage* for the Deduplication Repository data disks. select the object storage account that you had created with the “*Use for Deduplication Repository*” option. Select the physical devices reserved for Deduplication that will be used for the deduplication index and folder and click *Create* to enable the Deduplication Repository.
  - Enable iSCSI target mode to enable host clients running backup software to communicate with StorSafe via the iSCSI protocol.
- Select the *Virtual Libraries* tab and click the “+” icon to create virtual tape libraries with IBM drives:
  - For IBM i host clients, select the virtual tape library type as *FalconStor FALCON TS3500L32 (03584L32)* or *FALCON TS3500L32 (03584L32)* and the media type as *ULTRIUM3 (LT03)* or newer. By using Falcon library types, you get the 3584-403 device types to configure on IBM i host clients.
  - Set the library name, the drive name prefix, number of drives, barcode range for tapes, and set the number of Import/Export slots to 1.
  - Enable Tape Capacity On demand (COD) to create small resources for your tapes and then automatically allocate additional space when needed. The minimum value for the incremental size is (Maximum Capacity – Initial Tape Size) / 63.
  - Leave default settings and do not enable any service options.
- Configure iSCSI from the client side according to the *Configure an iSCSI client* section below.
- Select the *Host Clients* tab and click the “+” icon to add host clients to which virtual tape libraries will be assigned for backup:
  - Specify the client name and select *iSCSI* as the communication protocol.
  - Click the *iSCSI* tab to select the initiators that this client will use. If the initiator does not appear, you may need to rescan by clicking *Rescan Initiators*. The client iSCSI initiator name is the same as the one configured on the client side, for example, *iqn.1994-05.com.ibm:apacibmi74falcon*.
  - Do not enable CHAP, to allow unauthenticated access.
- Select the newly created host client to create an iSCSI target for the client. Click the *Manage* icon and select *Create iSCSI Target*. Enter the iSCSI target name as the one configured on the client side, for example, *iqn.2000-03.com.falconstor:h21-47.ibm94*. Select the IP address of the adapter on the StorSafe server. Leave the starting LUN as 0. One iSCSI target should be created for each iSCSI client initiator. Refer to “Configure an iSCSI client” for more information.
- Select the newly created host client to assign virtual tape libraries to the new iSCSI target. Click the *Assign* icon. Select available libraries to assign.
- On the client side, to discover assigned devices, run the IPL I/O processor option to send a login request. You can run the IPL using a SQL command or the Start System Service Tools command (STRSST):

SQL Command

- Run the SQL command with the IPL I/O processor option:  
`CALL QSYS2.CHANGE_IOP(IOP=>'ISCSI', OPTION=>'IPL');`

STRSST Command

- Run STRSST on the client to check the system bus resource and execute a bus reset to the iSCSI bus resource:  
I/O debug to 298A-001 IOP resource (option 6):

```

Logical Hardware Resources on System Bus

System bus(es) to work with . . . . . *ALL *ALL, *SPD, *PCI, 1-9999
Subset by . . . . . *ALL *ALL, *STG, *WS, *CMN, *CRP

Type options, press Enter.
2=Change detail 4=Remove 5=Display detail 6=I/O debug
7=Display system information
8=Associated packaging resource(s) 9=Resources associated with IOP

Opt Description Type-Model Status Resource
_ Virtual System Bus - Operational LB02
6 Virtual IOP 298A-001 Operational CMB02

```

Run the IPL I/O processor option that will send a login request from the client iSCSI initiator to the StorSafe server with a nonexistent target:

```

4. IPL I/O processor

```

If everything is successful, the StorSafe server receives the iSCSI login request with the following sample messages in the system log:

```

May 30 14:34:23 h21-47 fsiscsid[9033]: IPSTOR||1653892463||I||0x0000c351||Login
to the target %1 from the initiator %2||iqn.2000-03.com.falconstor:h21-
47.ibm194||iqn.1994-05.com.ibm:apacibmi74falcon
May 30 14:34:23 h21-47 kernel: FSISCSI client 14 initiator iqn.1994-
05.com.ibm:apacibmi74falcon type 2 login request to target iqn.2000-
03.com.falconstor:h21-47.ibm194 from 172.24.2.94, conn 4020293, new tcp session
May 30 14:34:23 h21-47 kernel: FSISCSI conn 4020293 create new session 62603.
May 30 14:34:24 h21-47 kernel: svdp_get_cpu: vdev 536 cpu 31.
May 30 14:34:24 h21-47 kernel: svdp_get_cpu: vdev 537 cpu 32.
May 30 14:34:25 h21-47 kernel: svdp_get_cpu: vdev 538 cpu 33.
May 30 14:34:25 h21-47 kernel: [vtl_tde_537|4018142] TLE_INFO: VDrive 537
bPowerOnReset is set to 1, CDB[0]=0h vtape=-1 [n/a]
May 30 14:34:25 h21-47 kernel: IOCORE1 [kworker/31:1|3523459] release_vdev,
releasing vdev 536 without a reservation
May 30 14:34:26 h21-47 kernel: [vtl_tde_538|4018147] TLE_INFO: VDrive 538
bPowerOnReset is set to 1, CDB[0]=0h vtape=-1 [n/a]

```

- Confirm the iSCSI device is now available to the client. For example, the 3584-403 is displayed as a FalconStor vendor device Type-Model configured for the client:

|   |              |          |             |          |
|---|--------------|----------|-------------|----------|
| — | Tape Library | 3584-403 | Operational | TAPMLB03 |
| — | Tape Unit    | 3580-006 | Operational | TAP05    |
| — | Tape Unit    | 3580-006 | Operational | TAP06    |

- Select the *Deduplication* tab to use the default deduplication policy or create a new policy. For a new policy, enter a name and select the *Inline* deduplication trigger. Leave default priority and retry settings.
- If applicable, on the primary StorSafe server, designate the replica server as the target server and create deduplication/replication policies.
- On the IBM i host client, start your backup software to write data on virtual tapes.



## Configure an iSCSI client

This section provides an example of configuring an IBM i host client with StorSafe via iSCSI. You need to create a power server running IBM I and install IBM iSCSI packages and required PTF files on that server.

One iSCSI target is created in StorSafe for each iSCSI client initiator. The StorSafe PowerVS instance is running Ethernet under an IBM Cloud private network that belongs to a VLAN subnet; no extra VLAN is set on the StorSafe server.

### Check Type-Model

Confirm the IBM PTF is installed on the client by checking the *Type-Model*, which should display as 298A-001. This indicates that the iSCSI bus IOP resource is operational on the client. If it does not exist, contact IBM to install the required PTF file.

| Opt | Description | Type-Model | Status      | Resource Name |
|-----|-------------|------------|-------------|---------------|
|     | Virtual IOP | 298A-001   | Operational | CMB01         |

### Create an iSCSI target

For IBM i version 7.3 or higher, use the IBM Navigator for i GUI to create an iSCSI target. Select *System Services* → *iSCSI* Tab → *Action* → *Create iSCSI Target* → Enter target iSCSI Qualified Name (IQN), target IP address, or hostname.

For older IBM i versions, use the SQL service commands as described below.

### SQL commands

1. Confirm the SQL service is operational by running the run `STRSQL` command. This service is used for configuring communication between the client iSCSI initiator and StorSafe iSCSI target.

```

Enter SQL Statements

Type SQL statement, press Enter.
Session was saved and started again.
Current connection is to relational database APACIBMI.
Session was saved and started again.
Current connection is to relational database APACIBMI.
> CALL QSYS2.ADD_ISCSI_TARGET(
    TARGET_NAME=>'iqn.2000-03.com.falconstor:h2-70.ibm194',
    TARGET_HOST_NAME=>'172.24.2.70',
    INITIATOR_NAME=>'iqn.1994-05.com.ibm:apacibmi74falcon'
)
CALL statement complete.
Session was saved and started again.
Current connection is to relational database APACIBMI.
==>

```

2. Run the SQL command to add the iSCSI target and the initiator information on the client; you can set the target to any value matching IQN patterns, for example:

```
CALL QSYS2.ADD_ISCSI_TARGET(
TARGET_NAME=>'iqn.2000-03.com.falconstor:h21-47.ibm194',
TARGET_HOST_NAME=>'172.22.21.47',
INITIATOR_NAME=>'iqn.1994-05.com.ibm:apacibmi74falcon');
```

The iSCSI target name does not exist on StorSafe yet. After the SQL command completes, you will use the same target name on StorSafe, when configuring the iSCSI client.

3. Run the SQL command with the IPL I/O processor option that will send a login request from the client iSCSI initiator to the StorSafe server with a nonexistent target:

```
CALL QSYS2.CHANGE_IOP(IOP=>'ISCSI', OPTION=>'IPL');
```

4. Confirm the IPL I/O processor successfully completes by checking the StorSafe system log, /var/log/messages. Make a note of the target name in the log; you will need to use it when configuring the iSCSI target in StorSight. You will see a message similar to the following:

```
May 30 14:31:36 h21-47 fsiscsid[9033]:
IPSTOR||1653892296||E||0x0000c352||Login request to nonexistent target %1
from initiator %2||iqn.2000-03.com.falconstor:h21-47.ibm94 (ip
172.22.21.47)||iqn.1994-05.com.ibm:apacibmi74falcon
```

# Add Server Resources

This section describes the configuration changes you can make to add system resources to an operational StorSafe server.

## Expand object storage for an on-premises system

1. Complete the FalconStor online form at <http://ibmexpansion.falconstor.com/> to get a new license keycode for the new required space.
2. From the StorSight portal, select *Administration* -> *StorSight*. Click *License*, and the “+” icon to add your new license keycode received from FalconStor. Register your license.
3. In the *Manage* tab, click the *Settings* tab, click the *Deduplication Repository* icon, click the *Manage* icon, and select *Add Deduplication Data Storage*. Specify the expansion size of object storage in GB. You must have enough object storage capacity; the system cannot check whether there is sufficient free space.
4. Check sizing guidelines to see if you also need additional system resources, such as CPU cores, memory, or block storage.

## Expand object storage for a power system

Follow the steps below if you need to increase the size of COS for deduplication repository data:

1. From the IBM Cloud portal, select your power system from the *Resource list* -> *Services and software* menu.
2. Click *Virtual Appliances* and select your instance.
3. Click *Edit details* to increase the licensed repository capacity of object storage. Check the box next to “*I agree to the Terms and conditions*” and then click *Save edits and order*. You will receive an email from [noreply@falconstor.com](mailto:noreply@falconstor.com). Check your spam folder if you do not see the email in your inbox within one hour. Click the link in the e-mail to fill in a web form. You will receive another e-mail from FalconStor to confirm your capacity order. A new capacity license key will be delivered to you by FalconStor Customer Support.
4. From the StorSight portal, select *Administration* -> *StorSight*. Click *License*, and the “+” icon to add your new license keycode received from FalconStor. Register your license.
5. In the *Manage* tab, click the *Settings* tab, click the *Deduplication Repository* icon, click the *Manage* icon, and select *Add Deduplication Data Storage*. Specify the expansion size of object storage in GB. You must have enough object storage capacity; the system cannot check whether there is sufficient free space.
6. Check sizing guidelines to see if you also need additional system resources, such as CPU cores, memory, or block storage. If applicable, follow related steps in this section.

## Add block storage for a power system

Follow the steps below if you need to attach additional block storage to a StorSafe server:

1. From the IBM Cloud portal, select your power system from the *Resource list* -> *Services and software* menu.
2. Click *Virtual Appliances* and select your instance.
3. Click *Create volume* for a new volume or click *Attach existing* for an existing volume.
4. Set values for related parameters. Acknowledge and submit changes.
5. From the StorSight portal, select *Rescan* on the *Physical Devices* tab to detect new devices.

## Add memory for a power system

Follow the steps below if you need to change the amount of memory of a StorSafe instance. For example, when you increase the size of object storage for the deduplication repository, you may also need to increase memory to be used for deduplication.

1. From the IBM Cloud portal, select your power system from the *Resource list* -> *Services and software* menu.
2. Click *Virtual Appliances* and select your instance.
3. Stop StorSafe services and shut down the instance.
4. Click *Edit details* to enter a new value for memory.  
For details, refer to the following IBM document “Modifying a Power Systems Virtual Server instance” <https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-modifying-server>.
5. Power on the instance and restart StorSafe services for the changes to take effect.

## Add CPU for a power system

Follow the steps below if you need to change the number of CPU cores of a StorSafe instance. For example, when you increase the size of object storage for the deduplication repository, you may also need to increase the number of CPU cores.

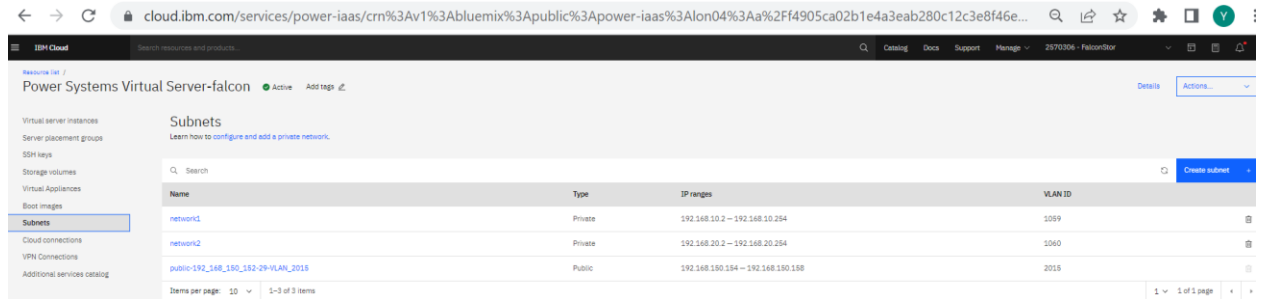
1. From the IBM Cloud portal, select your power system from the *Resource list* -> *Services and software* menu.
2. Click *Virtual Appliances* and select your instance.
3. Make sure it is in an *Active* or *Stopped* state, and click *Edit details* to enter new values for CPU cores.  
For details, refer to the following IBM document “Modifying a Power Systems Virtual Server instance” <https://cloud.ibm.com/docs/power-iaas?topic=power-iaas-modifying-server>
4. Restart the StorSafe services for the changes to take effect.

## Add network for a power system

### Private network

Follow the steps below if you need to add a private network interface to a StorSafe instance:

1. From the IBM Cloud portal, select your power system from the *Resource list* -> *Services and software* menu.
2. Select *Subnets* and click *Create subnet*.



3. Add subnets for your network.

**New subnet**

Name:

CIDR <sup>①</sup>:

Gateway:

IP ranges <sup>①</sup>:

DNS server:

Cloud Connection (optional):

2nd Cloud Connection for redundancy (optional):

4. Select your Power StorSafe instance.
5. Stop StorSafe services.

6. Click *Attach existing network* and select a network to add.

**Network interfaces**  
At least one interface, public or private, is required.

Public networks

☒ On

| Name    | IP address | External IP | Gateway | MAC address | VLAN ID | CIDR |
|---------|------------|-------------|---------|-------------|---------|------|
| public- |            |             |         |             | 2072    |      |

**Private networks**

Search

**Attach existing network**

| Name                            | IP address   | Gateway     | MAC address       | VLAN ID | CIDR                                  |
|---------------------------------|--------------|-------------|-------------------|---------|---------------------------------------|
| private-172.24.2.64/27-VLAN-661 | 172.24.2.71  | 172.24.2.65 | fa:d8:5e:6b:ac:22 | 661     | 172.24.2.64/27 <a href="#">Detach</a> |
| private-172.24.5.0/24-VLAN-889  | 172.24.5.217 | 172.24.5.1  | fa:d8:5e:6b:ac:21 | 889     | 172.24.5.0/24 <a href="#">Detach</a>  |

**Attach an existing network**

Existing networks

private-172.24.8.0/24-VLAN937

IP range

172.24.8.2 – 172.24.8.254

IP address

☒ Automatically assign IP address from IP range

☐ Manually specify an IP address from IP range

Specified IP address

Cancel Attach

7. Make sure network routers are configured properly for the traffic in your network infrastructure.

8. Open a Linux shell on the StorSafe server and type the following commands to complete the network device configuration:
  - a. `# systemctl restart NetworkManager` (to restart the network)
  - b. `# ls /sys/class/net` (to see the new interface device `ethn`, for example `eth3`)
  - c. `# cat /sys/class/net/eth3/address` (to get the HW MAC address of the new network device)
  - d. `# cd /etc/sysconfig/network-scripts`  
`# cp ifcfg-eth0 ifcfg-eth3` (to copy one network file, for example, `ifcfg-eth0`, to the new one, for example `ifcfg-eth3`)
  - e. `# vi ifcfg-eth3` (to set related parameters for the new interface file, such as the following)
    - i. `DEVICE=eth3`
    - ii. `HWADDR=`
    - iii. `IPADDR=`
    - iv. `NETMASK=`

### Public network

You can have only one public network. Follow the steps below if you need to add a public IP address for a StorSafe instance:

1. From the IBM Cloud portal, select your power system from the *Resource list -> Services and software* menu.
2. Click *Virtual Appliances* and select your instance.
3. Check the *Public networks* box.

| Network interfaces<br><small>At least one interface, public or private, is required.</small><br>Public networks<br><input checked="" type="checkbox"/> On |                 |                 |                 |                    |         |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|--------------------|---------|--------------------|
| Name                                                                                                                                                      | IP address      | External IP     | Gateway         | MAC address        | VLAN ID | CIDR               |
| public-192.168.150.152-29-VLAN_2015                                                                                                                       | 192.168.150.158 | 158.176.161.158 | 192.168.150.153 | fa:4c:9b:09:bcd:20 | 2015    | 192.168.150.152/29 |
| Private networks                                                                                                                                          |                 |                 |                 |                    |         |                    |

### Network considerations

1. You can only disable a public IP address while deduplication has not been enabled on the StorSafe server. After disabling the public IP address, you must restart StorSafe services. If deduplication is already enabled while the public interface is in effect, contact FalconStor Technical Support for help.
2. If you need to remove any network interfaces on the StorSafe server, contact FalconStor Technical Support for help.

# APPENDIX 1 -

## Measures for Security Threats

This section describes the security policy implemented for the underlying operating system (OS) and FalconStor StorSafe software product.

### OS packaging

The FalconStor software image contains a scaled down version of the Linux operating system, which contains only required packages. Only a subset of Linux support modules is included, to keep unneeded services from being available for malicious entry points. The hardening of OS packaging is a controlled process and does not use an automated update program such as `yum` in order to avoid adding or updating unnecessary files. At each product release, FalconStor makes sure the OS packages in the product USB image are up-to-date and do not contain any security vulnerabilities. If any major vulnerability is fixed in newer minor versions of the kernel used in the current USB, the USB kernel is also updated to that newer version. FalconStor will perform tests to ensure OS patches do not cause any issue to the software.

FalconStor regularly checks for vulnerabilities using the following methods:

- Vulnerability updates/newsletters from the Red Hat Security Response Team and Red Hat General Advisories web page for upcoming OS updates
- Security Content Automation Protocol (SCAP) reports
- Reports from vulnerability scanners such as Nessus® by Tenable Network Security from customers. FalconStor focuses on the 'High' and 'Medium' reported vulnerabilities to evaluate whether an OS patch is needed.

### OS security options

The following security options are enabled during installation:

- OS (`Grand Unified Bootloader`) GRUB security features allow setting a password so users cannot edit any grub entries or pass arguments to the kernel from the grub command line without entering the password.
- The Linux Audit framework can log system calls, such as, opening a file, killing a process, or creating a network connection. These audit logs can be used to monitor systems for suspicious activity.
- The Linux Advanced Intrusion Detection Environment (AIDE) can be configured with predefined rules to check the integrity of files and directories in the Linux operating system.
- The OpenSCAP scanner packages are included for the Security Content Automation Protocol (SCAP). SCAP content is based on the Security Technical Implementation Guide (STIG) published by the Department of Defense Cyber Exchange (DoD), which is sponsored by the Defense Information Systems Agency (DISA). It contains guidance on how to configure systems to defend against potential threats. The OpenSCAP scanner can be regularly run in order to apply required fixes and bring the system to a compliant state.

## **Authentication**

The following measures are taken in order to maintain a high level of security among services for authentication:

- Linux secure login with shadow passwords is used to access the server terminal console.
- Remote SSH access is disabled for the 'root' account on all StorSafe servers.
- A shared secret mechanism based on the Diffie-Hellman algorithm is used for authentication between:
  - Source and replica servers
  - Management portal and server
  - Host clients and server

The Diffie-Hellman key exchange sets a shared secret of 48 bytes between primary and target components. When a communication session starts, the primary authenticates itself with the target and generates two symmetric keys following the TLS 1.2 standard, one for sending and one for receiving data.

- The FalconStor StorSafe NAS feature provides two security modes to authenticate users/groups trying to access NAS shares:
  - Domain mode, where authentication is controlled by a Windows Active Directory Domain Controller; POSIX Access Control Lists (ACLs) can be set on files and folders.
  - User mode, where authentication is controlled by passwords that are set for each Windows user.
- FalconStor StorSight uses the Spring framework, where the security module is a flexible and powerful authentication and access control framework to secure Spring-based Java web applications.
- FalconStor StorSight offers the option to use an Active Directory or LDAP server for authenticating and authorizing users.
- FalconStor StorSight offers the Multi-Factor Authentication (MFA) option. MFA is a multi-step account login process that requires users to enter more information than just a password. Along with the password, users will be asked to enter a verification code sent to their email in order to validate their identity.

## **Communication**

The following measures are taken in order to maintain a high level of security for communication between software modules:

- Most of the standard communication ports are disabled and only those required for FalconStor software are left open. Although you may temporarily open some ports during initial setup of the FalconStor appliance, such as the telnet port (23) and FTP ports (20 and 21), you should shut them down after your work is complete.
- Non-standard dedicated communication ports are used by the software modules for internal communication. The list of used ports is available in an appendix in the user guide.
- The management portal and host clients use a secured RPC link to communicate with FalconStor servers.
- The web-based StorSight GUI can use an SSL certificate for secure https communication with the StorSight management module.

Encryption provides an additional layer of security during replication by securing data transmission over open, public networks. Initial key distribution is accomplished using the authenticated Diffie-Hellman exchange protocol. Subsequent session keys are derived from the master shared secret, making it very secure. The available replication encryption methods are ARC4 (128-bit), AES (128-bit), and AES (256-bit).

AES encryption is compliant with Federal Information Processing Standard (FIPS) 140-2; all cryptographic code/algorithms are located in a single FIPS 140-2 compliant software module.

The Mutual CHAP level of security allows the target and the initiator to authenticate to each other. A separate secret is set for each target and for each initiator in the storage area network (SAN).

The system checks whether the password meets the following complexity requirements in the default configuration:

- The password contains at least 14 characters
- The password meets at least two of the following conditions:
  - Contains at least one lower-case letter
  - Contains at least one upper-case letter
  - Contains at least one digit
  - Contains at least one space or one of the following special characters:  
[mailto:~!@#\$\$%^&\*()-\_+=~\|[{];:","<.>/]~!@#\$\$%^&\*()-\_+=~\|[{];:","<.>/?
- The password is not the same as the account or its reverse order.
- There are at least 5 different characters between an old and new password.

- The system locks the account if the user enters a wrong password for a number of times more than the threshold specified during product configuration; the default value is three times.
- The system allows setting the account locking duration for user accounts locked due to more than  $n$  login attempts with wrong passwords. The recommended locking duration is five minutes.
- When the account locking duration elapses, the account is unlocked automatically. The security administrator can also unlock the user accounts manually.
- When an account is locked, only the security administrator can unlock the account manually.

- Passwords are encrypted with AES 256.
- The password must be entered in encrypted text. That is, the entered password is presented by asterisks \* on the user interface. The password cannot be displayed in plain text in terminals or logs.
- A password without encryption in memory (for example, at login) must be overwritten right after being used.
- The password cannot be saved in log files, configuration files, cookies, or buffers without encryption.

- Access control is implemented for password files and common users cannot read or copy the encrypted content.
- A password in a text box cannot be copied.
- The old password is required for a password change.
- Users (except for administrators) cannot change the passwords of other user accounts.
- GUI console login authentication uses SHA 512.

### ***SNMP traffic***

SNMP user authentication uses the MD5 or SHA algorithm.

SNMP data traffic uses AES or DES for encryption of data sent over the network. A passphrase (8-127 characters) is used.

## **Data security**

### ***Data encryption***

Encryption can be enabled for:

- Virtual libraries to encrypt virtual tape data using an encryption key defined by the user
- Migrating tape data to an object storage account in-flight and at-rest (end-to-end) using an encryption key obtained for each tape from the Network Security Service (NSS) internal key management system
- Deduplication repository using an internal encryption key

### ***Data structure***

All data are stored as disaggregated without a file system construct; data layout is not discoverable outside the FalconStor server or across user accounts to provide the first “air-gap” security layer.

### ***WORM tapes***

On a virtual tape library or drive, the Write-Once-Read-Many (WORM) property can be enabled for tapes that support ULTRIUM5 media type and above. WORM tapes cannot be overwritten. WORM allows non-rewriteable and non-erasable data to be written and provides extra data security by prohibiting accidental data erasure. Since tapes are written once, they cannot be altered or overwritten by some virus/ransomware/other malicious software.

### ***Immutable Cloud Object Storage***

The immutable option can be enabled for tape migration to IBM Cloud Object Storage (COS). Immutable object storage locks data to provide a safe backup and to maintain data integrity. You can enable an object lock at the bucket level to get secure backups and long-term data retention. Retention policies ensure that data is stored in a non-erasable and non-rewritable manner for a specified time frame. Data cannot be changed until the retention period has expired. Once the retention period is over, data can be unlocked for further actions, according to your company policies.

From the IBM COS management GUI, you can configure the retention policy of your bucket and set the minimum and default values to zero, and the maximum retention period to the number of days you want to lock the bucket data.

From the FalconStor portal, you can set a tape migration policy for a virtual tape library or drive. At the end of each backup job, the tape is ejected to the vault, where the tape data gets packaged and migrated to the object storage of a cloud provider.

This copy of data remains intact during the retention period and is protected from malicious activity or accidental deletion.

### ***Tape shredding***

Just as deleting a file from your hard drive does not completely destroy the file, deleting a virtual tape does not completely destroy the data on the tape. If you want to ensure that the data is unrecoverable, you must shred the tape. Shredding a virtual tape destroys all data on the tape, making it impossible to recover the data. Tape shredding uses a military standard to destroy data on virtual tapes by overwriting it with a random pattern of bits, rendering the data unreadable.

### ***Data Isolation for multi-tenancy***

StorSight provides a secure multi-tenant architecture that allows Managed Service Providers (MSPs) and large enterprises to offer data protection as a service to their customers, called tenants. Each tenant (a business or organization) has its own secure computing environment, called a domain (i.e., *companyA.com*). While all storage and network resources are shared among domains, StorSight logically isolates data by restricting visibility of storage resources to specific customers. Therefore, customers only see their own data and are not aware of other tenants.

The landlord in a multi-tenant environment is a Super Administrator, the master role that does not belong to any domain. By default, there is one Super Administrator account per StorSight server; however, the Super Administrator can create additional Super Administrator accounts.

The Super Administrator creates an administrator account for each company or organization, associating each tenant's domain with the account.

The Administrator for each domain then creates user accounts within their organization. Typically, the Administrator will create Administrator accounts for departments within their organization, possibly creating one Administrator for each department. Each Administrator will then create other user accounts as needed for their department.

For privacy, access to downstream storage can be isolated via storage pools, access to Fibre Channel targets via separate Fibre Channel zones, and access to networks via separate private subnets.

StorSafe Servers can be assigned to a customer domain in exclusive or shared mode:

- In exclusive mode, the server is only available to the users in that customer domain.
- In shared mode, the server is available to users of multiple customers. Data isolation among those customers is by assigning different storage pools to each single customer so they can only see resources created from storage pools assigned to them.

A hierarchy of user account types is used to control access to the system:

- **Superadmin:** A master role that does not belong to any domain with full access to StorSight who can create customer domains, add and assign StorSafe servers, virtualize physical resources, reserves resources for specific usages (configuration, tapes, deduplication, NAS), create storage pools of virtualized storage, perform global configuration and maintenance of servers, configure failover, and create the single instance deduplication repository.
- **Admin:** The administrator for a customer domain who can create User or Viewer accounts and assign these accounts to specific storage pools, virtual resources, and clients. An Administrator can also create and manage virtual resources for their domain.  
An Administrator of a server in exclusive mode has the same rights as the superadmin (root) user on the server.
- **User:** A tenant user with read-write access to resources that have been created by that user or assigned to that user by the domain admin.
- **Viewer:** A tenant user with read-only access to resources within a domain who can view configuration and reports as well as receive alerts but cannot make any configuration changes.

Accounts with a higher or equal role can change the properties of other accounts. For example, Superadmin can change the password of other Superadmin, Admin, User, or Viewer accounts; Admin can change the password of other Admin, User, and Viewer accounts. Users and Viewers can only change their own passwords.

### ***Event logging***

- The Linux Audit framework can log system calls, such as, opening a file, killing a process or creating a network connection. These audit logs can be used to monitor systems for suspicious activity.
- The Linux Advanced Intrusion Detection Environment (AIDE) can be configured with predefined rules to check the integrity of files and directories in the Linux operating system.
- Any operation performed via the management portal or command line interface that changes the current state/configuration is recorded in the event log.
- Any user login and logout are recorded in the event log.

### ***General Security Guidelines***

FalconStor recommends the following general guidelines to ensure security:

- Place FalconStor appliances in a secure location protected by firewalls and accessible only by trusted people since these appliances are service units and not general-purpose computers.
- Do not create any shares on appliances.
- Do not install any unauthorized software.
- Do not open any unnecessary communication ports.
- Apply only OS patches certified by FalconStor.

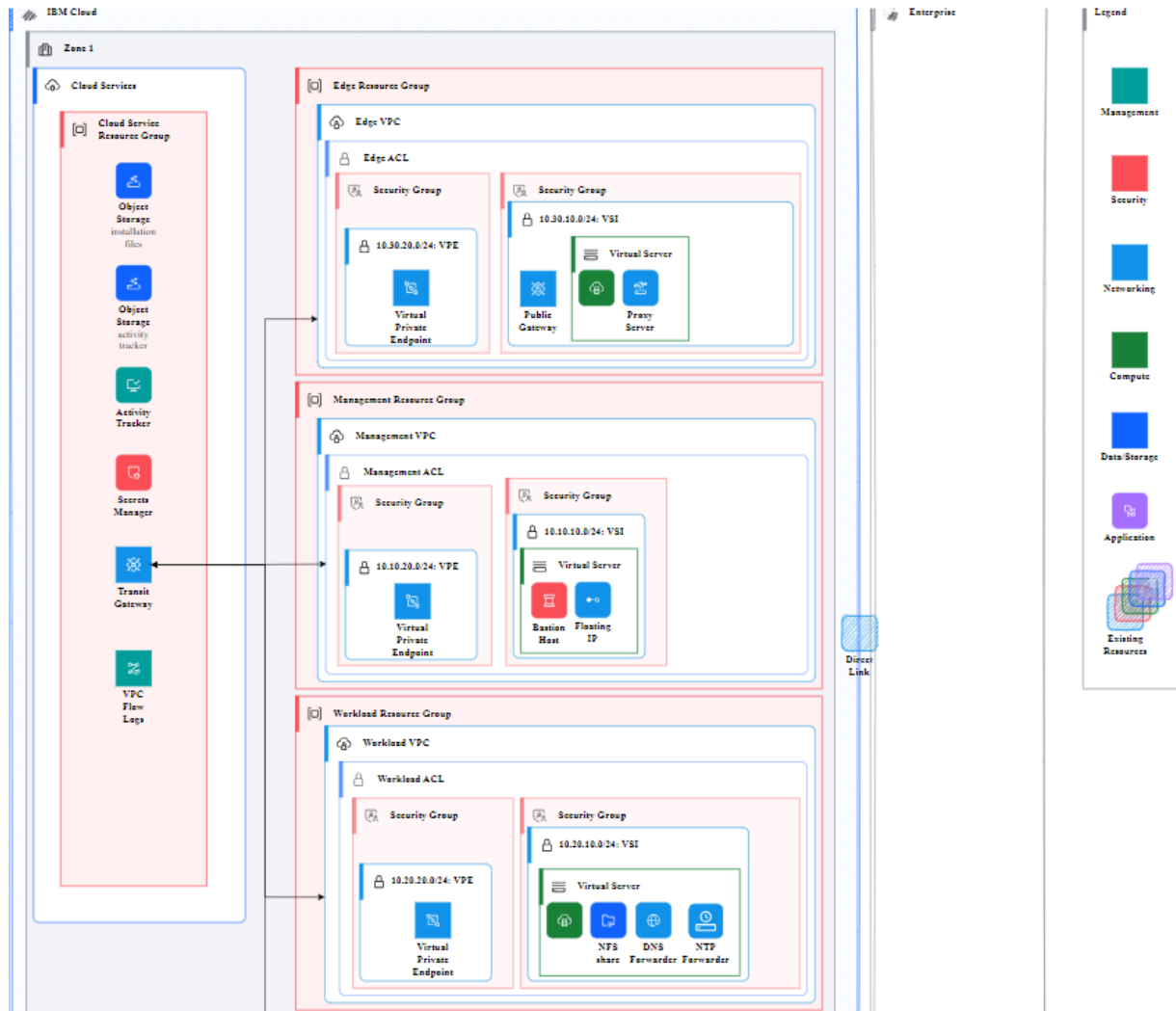
## ***APPENDIX 2 -***

# ***IBM Deployable Architecture Workspace***

This section describes the steps to use the IBM Deployable Architecture from the catalog to build a Power Systems Virtual Server (PowerVS) workspace with a Virtual Private Cloud (VPC) landing zone. Then, you can select the CRN of this workspace when you deploy a FalconStor StorSafe tile.

The deployable architecture for PowerVS workspace provides:

- Three separate VPCs for service isolation
- Three Intel Virtual Server Instances (VSI) in the classic environment:
  - Server running DNS, NTP, and NFS services
  - Server running Squid proxy server to access the public Internet
  - Server with a floating IP address acting as a jump box to access other servers
- Two cloud connections for a transient gateway to access the classic environment
- Two networks, one for management and one for backup
- Two Cloud Object Storage (COS), one for management and one for workload

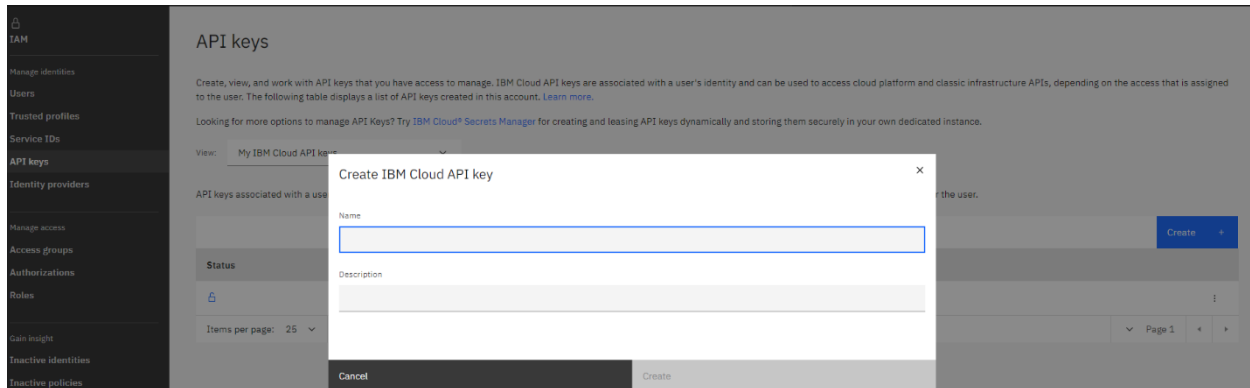
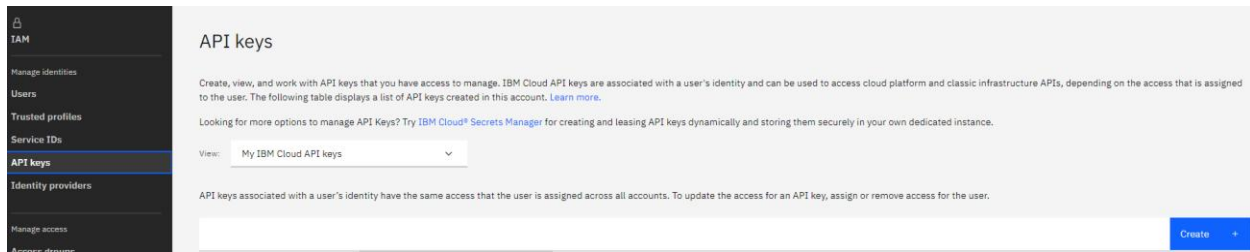


## Prerequisites

1. Create your API key.
2. Generate your SSH private and public keys on your machine that will access resources in the workspace.
3. Add IBM Secrets Manager service that will hold your keys for centralized management.
4. Create an IBM project to manage code-based deployments across accounts, collaborate with team members, and maintain compliance.

### Create your API key

Select Access (IAM) in the menu bar Manage option. Select *API keys* in the left panel, click *Create* and give a name to your key.



### Generate your SSH keys

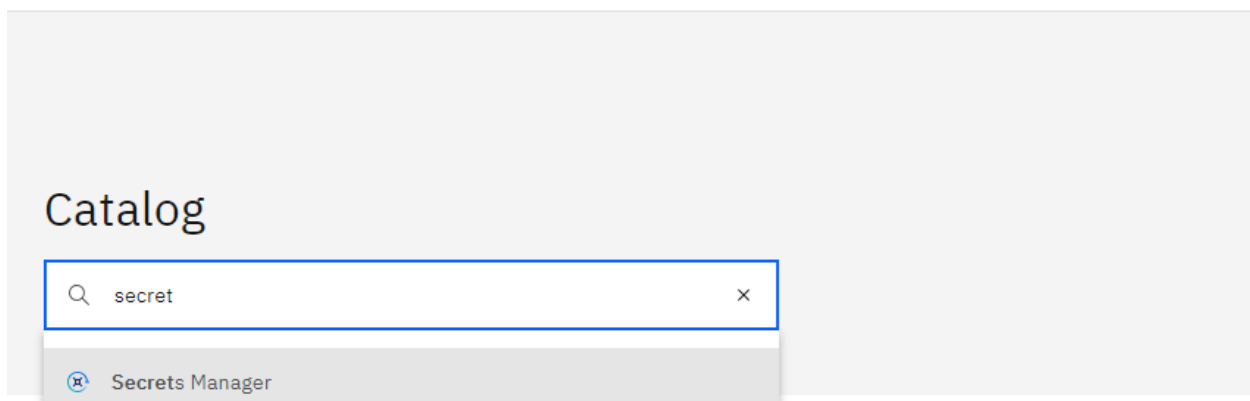
From the machine you want to access resources in the workspace, run the following command to generate an SSH public key in `.ssh/id_rsa.pub` and a private key in `.ssh/id_rsa`:

```
ssh-keygen
```

### Add IBM Secrets Manager service

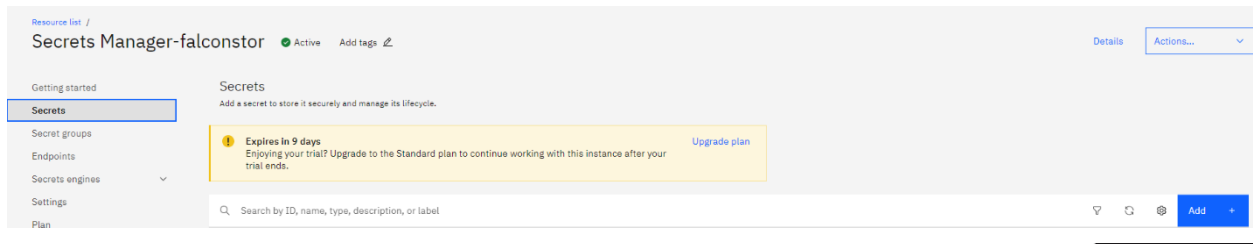
You can create one Secrets Manager service for your company or create several per tenant in a multi-tenancy environment.

In the IBM catalog, type 'secrets' and select *Secrets Manager*. You can use all default options. For clarity, create three groups for secret keys, for example, '*apikey*', '*private\_ssh*', '*public\_ssh*'.



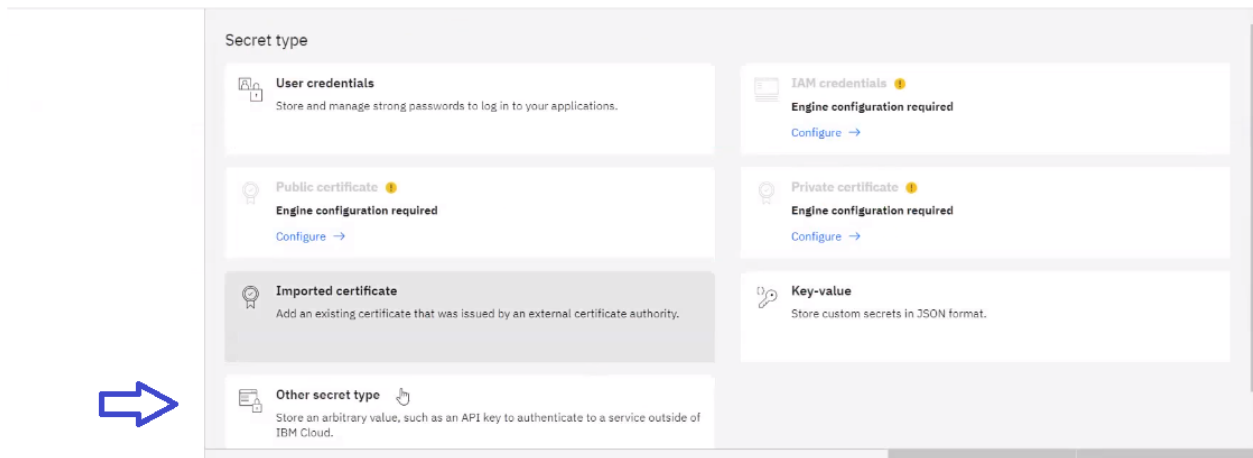
Go back to your API key and copy the key.

Select your Secrets Manager in the *Security* section of the Resource list, select *Secrets* in the left panel, and click *Add* to add your keys to the Secrets Manager. Then, select *Other secret type*.



### Add a secret

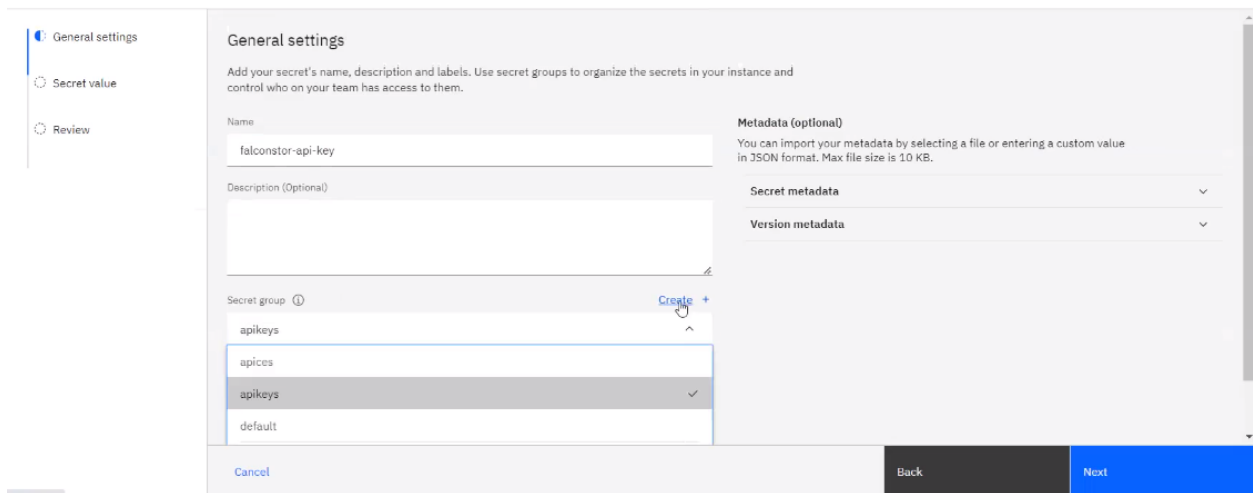
Choose a secret type to create. [Learn more](#)



Select the *apikey*s group and paste your copied API key.

### Add an arbitrary secret

Store an arbitrary or custom secret that you can use inside or outside of IBM Cloud. [Learn more](#)



## Add an arbitrary secret

Store an arbitrary or custom secret that you can use inside or outside of IBM Cloud. [Learn more](#)

In the same way, add your SSH public and private keys. Copy contents of `.ssh/id_rsa.pub` for the public key and `.ssh/id_rsa` for the private key. When copying the private key, you will need to enter a line on the top with `<< EOF` and a line at the end with `EOF` and a line break.

## Add an arbitrary secret

Store an arbitrary or custom secret that you can use inside or outside of IBM Cloud. [Learn more](#)

**Add an arbitrary secret**

Store an arbitrary or custom secret that you can use inside or outside of IBM Cloud. [Learn more](#)

General settings

Secret value

You can import your secret data by selecting a file or entering a custom value. Arbitrary secrets supports only text-based payloads. If you select a file, the service uses base64 encoding to store the data in your instance.

Learn more about base64 encoding

Enter data

Select file

```
<<-EOF
-----BEGIN OPENSSH PRIVATE KEY-----
b3BibnNzaC1rZXh0dEFAAAAABG5ybmlUAAAAEbm9uZQAAAAAAAAABAABlwAAAAdzc
2gicn
NkAAAAAuwFAAAAYFAAAAAAFAAAAAAFAAAAAAFAAAAAAFAAAAAAFAAAAAAFAAAAAAFA
-----END OPENSSH PRIVATE KEY-----
EOF
```

Max file size is 1 MB.

Set expiration date

Cancel Back Next

**Add an arbitrary secret**

Store an arbitrary or custom secret that you can use inside or outside of IBM Cloud. [Learn more](#)

General settings

Secret value

You can import your secret data by selecting a file or entering a custom value. Arbitrary secrets supports only text-based payloads. If you select a file, the service uses base64 encoding to store the data in your instance.

Learn more about base64 encoding

Enter data

Select file

```
MmAGNh
A9v93N3s9lVKF5AAAAE3Jvb3RAWVVWVhZ3MTRjgRk1TVkEBAgMEBQYH
-----END OPENSSH PRIVATE KEY-----
EOF
```

Max file size is 1 MB.

Set expiration date

Cancel Back Next

## Create an IBM Project

Select *Projects* in the *Resource* list to create a project to manage your deployments across accounts.

Create a project

+ Create

Define project details

Name

Name this project

Description (optional)

What's the purpose of this project?

Region

Dallas

Resource group

Default

Schematics workspaces and other project data are stored in the selected region and resource group. [Learn more.](#)

## Create a Deployable Architecture workspace

From the IBM catalog, select the *Power Virtual Server with VPC landing zone* tile. Then, select the *PowerVS workspace* option. Click *Review deployment options* in the right panel.

The screenshot displays the IBM Deployable Architecture Workspace interface. At the top, the 'Catalog' section shows a search bar and a list of categories. The 'Power Virtual Server with VPC landing zone' tile is selected. The product details page shows the 'Overview' and 'Architecture' sections. In the 'Variation' section, the 'PowerVS Workspace' option is selected, starting at \$486.09\*. The right panel shows the 'Summary' section with the total estimated cost of \$486.09/mo. A blue arrow points to the 'Review deployment options' button in the bottom right panel.

**Catalog**

Search the catalog...

Category: Viewing 234 products

Relevance

Internal IBM pricing displayed  
Classic infrastructure services might reflect internal IBM pricing, which should not be shared with external clients. Log out of your internal IBM Cloud account to view external pricing details.

**Power Virtual Server with VPC landing zone**  
By IBM  
Build a IBM Cloud® Power Virtual Servers (PowerVS) offering according to the best practices and requirements from the IBM Cloud.

**Cloud Object Storage on VPC for SAP HANA Backup**  
By IBM  
Deploy the capability to backup an SAP HANA database to IBM Cloud Object Storage in a single step.

**DevSecOps Application Lifecycle Management**  
By IBM  
Provides a set of predefined continuous integration, continuous deployment, and continuous compliance toolchain...

**Power Virtual Server for SAP HANA**  
By IBM  
Deploy SAP systems on Power infrastructure for deployable architectures.

**Red Hat OpenShift Container Platform on VPC landing zone**  
By IBM  
Creates Red Hat OpenShift workload clusters on a secure VPC network.

**Power Virtual Server with VPC landing zone**  
Build a IBM Cloud® Power Virtual Servers (PowerVS) offering according to the best practices and requirements from the IBM Cloud.

**Overview**  
IBM Cloud® Power Virtual Servers (PowerVS) is a public cloud offering that an enterprise can use to establish its own private IBM Power computing environment on shared public cloud infrastructure. PowerVS is logically isolated from all other public cloud tenants and infrastructure components, creating a private, secure place on the public cloud. This deployable architecture

**Architecture**  
Product version: v3.0.0  
How do you want to build this architecture?: Create a new architecture  
Version last updated: 11/03/2023

**Variation**  
PowerVS Workspace  
Starting at \$486.09\*  
See more  
PowerVS Quickstart  
Starting at \$681.78\*  
See more

**Summary**  
Power Virtual Server with VPC landing zone  
1 modules, 25 catalog resources  
Power Virtual Server with VPC landing zone  
25 resources  
\$486.09

**Architecture**  
Product version: v3.0.0  
How do you want to build this architecture?: Create a new architecture  
Version last updated: 11/03/2023

**Variation**  
PowerVS Workspace  
Starting at \$486.09\*  
See more  
PowerVS Quickstart  
Starting at \$681.78\*  
See more

\*Starting cost is an estimate based on available data. It does not include all resources, usage, licenses, fees, discounts, or taxes.

**Architecture overview** Permissions Security & compliance Help

**Total estimated cost** \$486.09/mo

**Review deployment options**

Click *Add to project* and select your existing project.

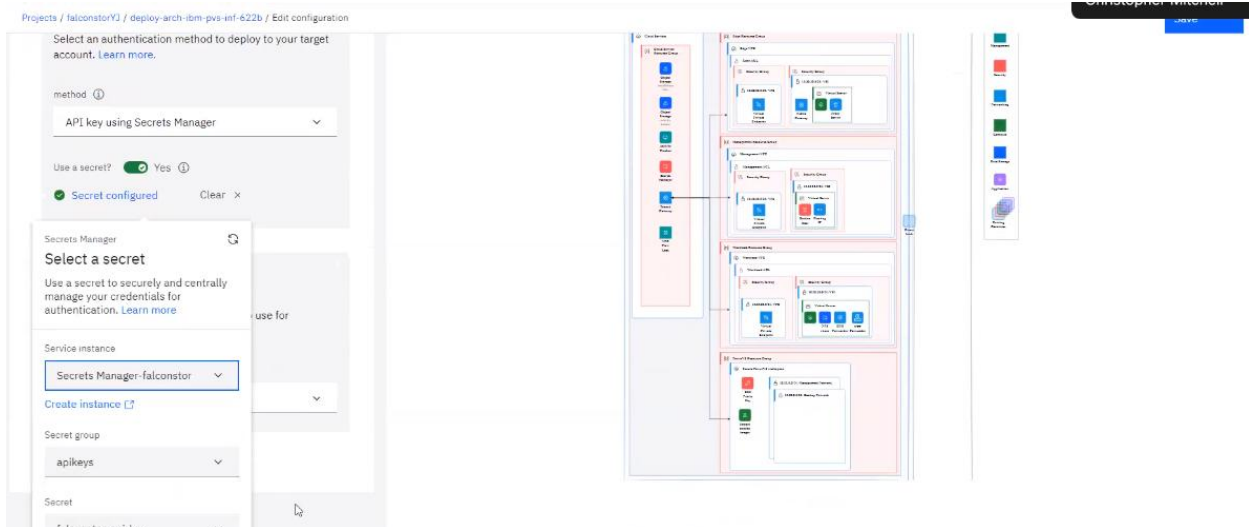
The screenshot shows the 'Add to project' dialog in the IBM Deployable Architecture Workspace. The dialog has three main sections: 'Add to project' (with a blue button), 'Create from the CLI' (with a terminal icon), and 'Work with code' (with a code icon). A sidebar on the right displays a 'Summary' of the architecture: 'Power Virtual Server with VPC landing zone', '1 modules, 25 catalog resources', and a cost of '\$486.09'.

Configure this instance of the deployable architecture. Enter a name for the configuration of the deployable architecture; include the deployment site region in the name for clarification.

Note that this workspace requires two cloud connections for the transient gateway to access the classic environment. Since there is a limit of two cloud connections per data center, make sure you select a site that does not have any cloud connections, so no direct link resource is used on that site.

The screenshot shows the 'Edit configuration' page in the IBM Deployable Architecture Workspace. The 'Define details' section has a 'Name' field with the value 'falconstor-dp-workspace-sydney04', which is highlighted by a blue arrow. Other fields include 'Based on' (Power Virtual Server with VPC landing zone), 'Version' (v3.0.0 (PowerVS Workspace)), and 'Environment (optional)' (None available). A 'Configure' button is at the bottom left. A sidebar on the right shows a summary of the architecture.

Scroll down to the *Security* section and select the authentication method as *API key* using the Secrets Manager. Select your Secrets Manager service and select the group of API keys, *apikey*s, in this example. Then select your API key. You need to click somewhere in the white space area to close the popup after entering keys.

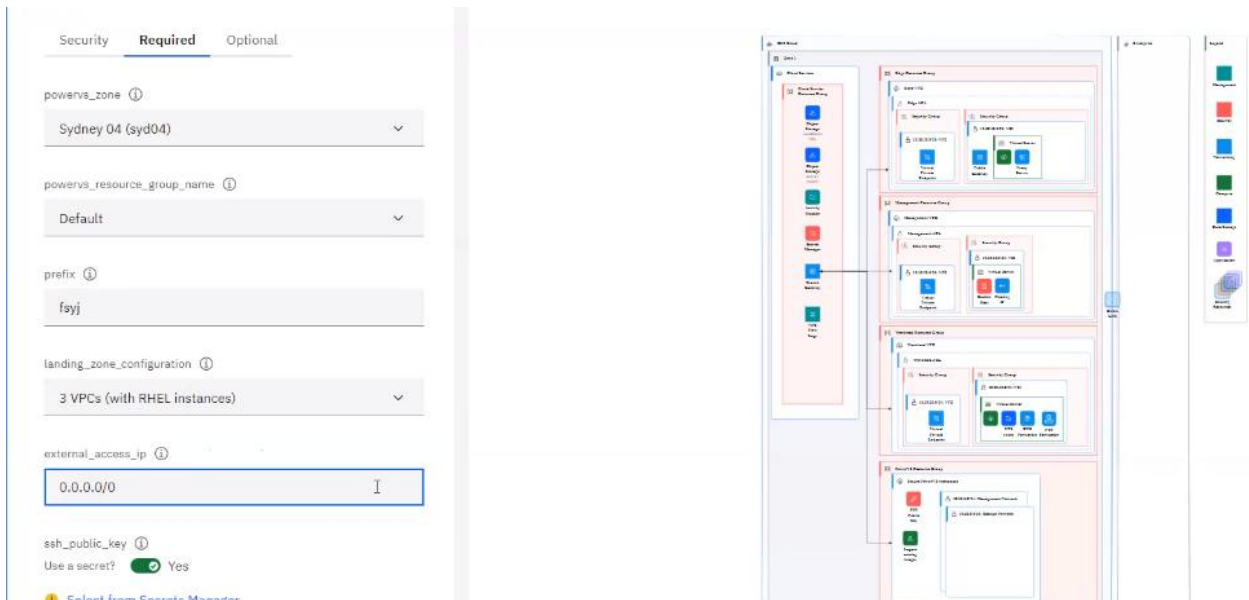


Click the *Required* tab and select a PowerVS region that does not have any cloud connections.

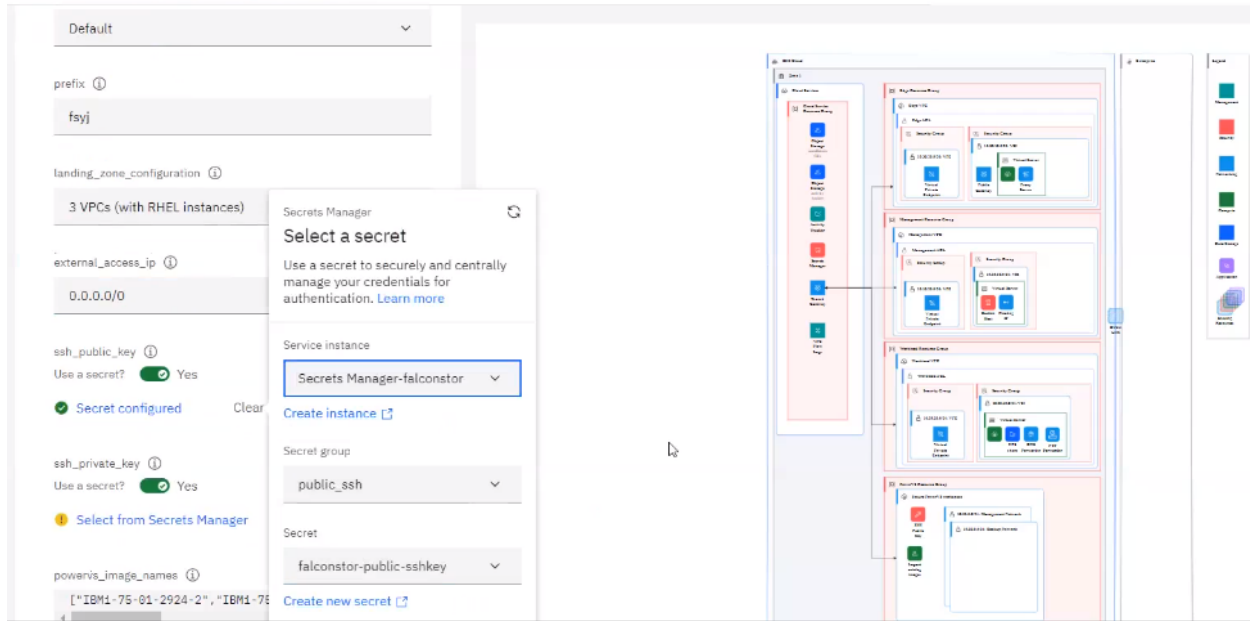
Enter a string in lowercase that will be used as a prefix for all resource names that will be created in this workspace. This allows you to easily list all these resources.

The three VSIs for DNS Server, NTP server, and NFS server can run RedHat or SUSE.

In the *external-access-ip*, enter a specific IP or a range of IP addresses that can access resources from outside. This can emulate a VPN that is not configured in the deployable architecture workspace. Enter 0.0.0.0/0 if you want all IP addresses to be able to access.

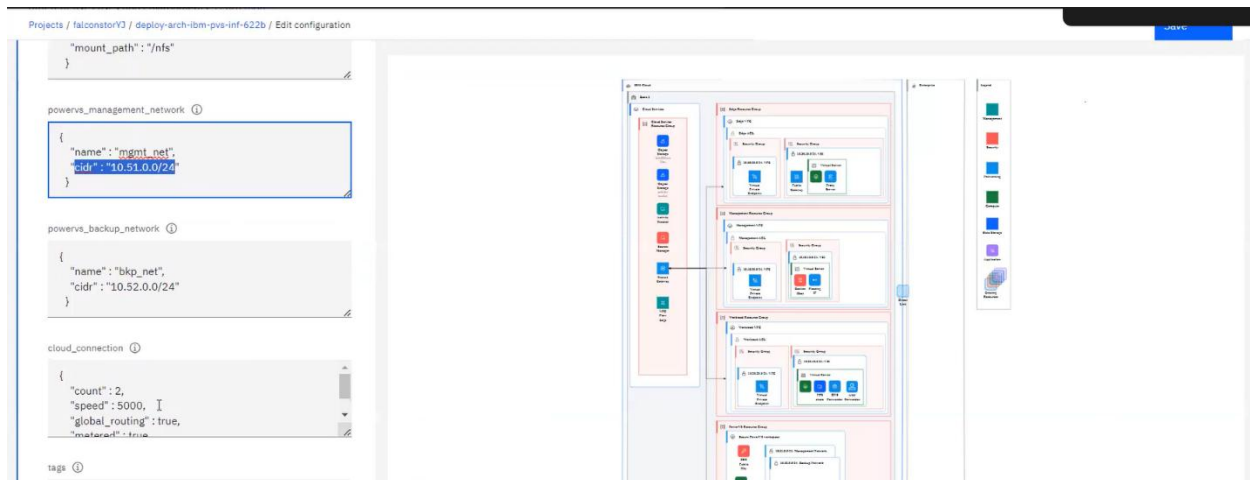


In a similar way to above, add your public and private keys.



Click the *Optional* tab to review default selections.

Check your network range for the two networks that can be used for management interface and backup application servers.



Click *Save* and then click *Validate* to start the terraform execution. It can take about 30 minutes to deploy terraform.

Once terraform execution is completed, click *Approve*, type 'I approve' and submit.

Validation successful ✕  
All changes are scanned for code errors, cost, and compliance.

Validation successful 2 minutes ago by yeggy.javadi@falconstor.com

Approval pending

I approve

Approve

Summary

**falconstor-dp-workspace-sydney04**  
Resource group: Default  
Location: Dallas  
1 modules, 24 catalog resources  
24 resources **\$339.62**  
Total estimated cost **\$339.62\*/mo**

Edit configuration

Deploy

## Output of Deployable architecture workspace

Click the deployable architecture workspace and check the results in the *Output* tab.

Projects / falconstor03 / falconstor-dp-workspace-sydney04 Edit

Deploying changes...

Resources **Outputs**

| Name                        | Value |
|-----------------------------|-------|
| access_host_or_ip           | —     |
| cloud_connection_count      | —     |
| dns_host_or_ip              | —     |
| nfs_host_or_ip_path         | —     |
| ntp_host_or_ip              | —     |
| powervs_backup_subnet       | —     |
| powervs_images              | —     |
| powervs_management_subnet   | —     |
| powervs_resource_group_name | —     |

In the *Resource* list, enter your prefix to list all created resources in the workspace.

Resource list Create resource

| Name                            | Group                  | Location  | Product                                    | Status    | Tags      |
|---------------------------------|------------------------|-----------|--------------------------------------------|-----------|-----------|
| Compute (4 / 17)                |                        |           |                                            |           |           |
| fscy-inet-svs-1                 | fscy-slt-edge-rg       | Sydney 1  | Virtual Server for VPC                     | Running   | schema... |
| fscy-jump-box-1                 | fscy-slt-management-rg | Sydney 1  | Virtual Server for VPC                     | Running   | schema... |
| fscy-private-svs-1              | fscy-slt-workload-rg   | Sydney 1  | Virtual Server for VPC                     | Running   | schema... |
| fscy-syd04-power-workspace      | Default                | Sydney 04 | Workspace for Power Systems Virtual Ser... | Active    | schema... |
| Networking (9 / 59)             |                        |           |                                            |           |           |
| fscy-edge-public-gateway-zone-1 | fscy-slt-edge-rg       | Sydney 1  | Floating IP for VPC                        | Available | —         |
| fscy-edge-public-gateway-zone-1 | fscy-slt-edge-rg       | Sydney 1  | Public Gateway                             | Available | schema... |
| fscy-edge-vpc                   | fscy-slt-edge-rg       | Sydney    | Virtual Private Cloud                      | Available | schema... |
| fscy-jump-box-1-fip             | Default                | Sydney 1  | Floating IP for VPC                        | Available | schema... |
| fscy-management-cos             | fscy-slt-service-rg    | Sydney    | Virtual Private Endpoint for VPC           | Healthy   | schema... |
| fscy-management-vpc             | fscy-slt-management-rg | Sydney    | Virtual Private Cloud                      | Pending   | schema... |
| fscy-transit-gateway            | fscy-slt-service-rg    | Sydney    | Transit Gateway                            | Available | schema... |
| fscy-workload-cos               | fscy-slt-service-rg    | Sydney    | Virtual Private Endpoint for VPC           | Healthy   | schema... |
| fscy-workload-vpc               | fscy-slt-workload-rg   | Sydney    | Virtual Private Cloud                      | Pending   | schema... |
| Storage (2 / 25)                |                        |           |                                            |           |           |
| fscy-attracker-cos-6govm0u9     | fscy-slt-service-rg    | Global    | Cloud Object Storage                       | Active    | schema... |
| fscy-cos-6govm0u9               | fscy-slt-service-rg    | Global    | Cloud Object Storage                       | Active    | schema... |

Identify two network subnets created for management and backup.

Power Systems Virtual Server

Workspaces

fscy-syd04-power-workspace

Compute

Virtual server instances

Virtual appliances

Shared processor pools

SSH keys

Boot images

Networking

Subnets

Subnets

| Name     | Type    | IP ranges               | VLAN ID |
|----------|---------|-------------------------|---------|
| blkp_net | Private | 10.52.0.4 — 10.52.0.254 | 1533    |
| mgmt_net | Private | 10.51.0.4 — 10.51.0.254 | 878     |

Items per page: 10 1–2 of 2 items 1 1 of 1 page

Identify the SSH key created for secure remote connections.

cloud.ibm.com/power/ssh-keys

fsyj-syd04-power-workspace / SSH keys

Search resources and products...

fsyj

Create SSH key

| Name                   | Key     | Date created                    |
|------------------------|---------|---------------------------------|
| fsyj-syd04-pcs-ssh-key | ssh-rsa | November 9, 2023 at 11:50:27 AM |

Items per page: 10 1-1 of 1 item

Check VPC subnets.

Subnets for VPC

Region: Sydney Search items

Create

| Name                       | Status    | Resource group         | Virtual Private Cloud | Location | IP range      | Public gateway |
|----------------------------|-----------|------------------------|-----------------------|----------|---------------|----------------|
| fsyj-management-vpe-zone-1 | Available | fsyj-slz-management-rg | fsyj-management-vpc   | Sydney 1 | 10.10.20.0/24 | —              |
| fsyj-management-vsi-zone-1 | Available | fsyj-slz-management-rg | fsyj-management-vpc   | Sydney 1 | 10.10.10.0/24 | —              |
| fsyj-management-vpn-zone-1 | Available | fsyj-slz-management-rg | fsyj-management-vpc   | Sydney 1 | 10.10.30.0/24 | —              |
| fsyj-edge-vsi-zone-1       | Available | fsyj-slz-edge-rg       | fsyj-edge-vpc         | Sydney 1 | 10.30.10.0/24 | 159.23.88.211  |
| fsyj-workload-vsi-zone-1   | Available | fsyj-slz-workload-rg   | fsyj-workload-vpc     | Sydney 1 | 10.20.10.0/24 | —              |
| fsyj-edge-vpe-zone-1       | Available | fsyj-slz-edge-rg       | fsyj-edge-vpc         | Sydney 1 | 10.30.20.0/24 | —              |
| fsyj-workload-vpe-zone-1   | Available | fsyj-slz-workload-rg   | fsyj-workload-vpc     | Sydney 1 | 10.20.20.0/24 | —              |

Items per page: 10 1-7 of 7 items

Check cloud connections.

fsyj-syd04-power-workspace / Cloud connections

Cloud connections establish connections between PowerVS and other IBM Cloud resources. Create two connections to ensure a redundant connection exists. A maximum of two connections can be created.

Search Create connection

| Connection name | Connection ID                        | Speed  | IBM Cloud Transit Gateway | Status      |
|-----------------|--------------------------------------|--------|---------------------------|-------------|
| syd04-conn-1    | 82bd4acb-f5f0-4f8f-8d70-74c9b5e6f5dd | 5 Gbps | Enabled                   | Established |
| syd04-conn-2    | a37b46ab-c7d5-403b-bf6e-35aa9912f002 | 5 Gbps | Enabled                   | Established |

Check Cloud Object Storage.

Virtual private endpoint gateways for VPC

Region: Sydney Search items

Create

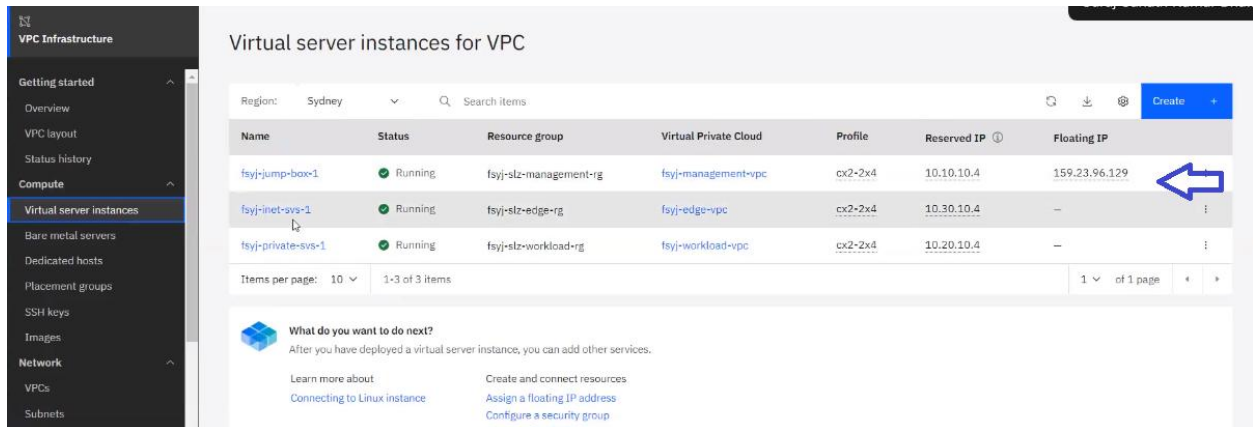
| Name                | Status | Resource group      | Service details      | Service endpoint                | Virtual Private Cloud | IP addresses |
|---------------------|--------|---------------------|----------------------|---------------------------------|-----------------------|--------------|
| fsyj-workload-cos   | Stable | fsyj-slz-service-rg | Cloud Object Storage | s3.direct.au-syd.clo... +1 more | fsyj-workload-vpc     | 10.20.20.4   |
| fsyj-management-cos | Stable | fsyj-slz-service-rg | Cloud Object Storage | s3.direct.au-syd.clo... +1 more | fsyj-management-vpc   | 10.10.20.4   |

Items per page: 10 1-2 of 2 items

## Access to other servers via jump box

To remotely access other servers via SSH, run the following command, where `access_host_or_ip` is the jump box IP address and `vpc_instance_ip` is the server IP that you want to access:

```
ssh -A -o ServerAliveInterval=60 -o ServerAliveCountMax=600 -o ProxyCommand="ssh -W %h:%p root@<access_host_or_ip>\\" root@<vpc_instance_ip>
```



In this example, *fsyj-jump-box-1* is the jump box host with a public floating IP, *fsyj-inet-svs-1* is the proxy server, and *fsyj-private-sys-1* is the DNS/NTP/NFS server.

For example, to access the DNS/NTP/NFS server with IP 10.20.10.4, run:

```
# ssh -A -o ServerAliveInterval=60 -o ServerAliveCountMax=600 -o ProxyCommand="ssh -W %h:%p root@159.23.96.129" root@10.20.10.4
```

```
Activate the web console with: systemctl enable --now cockpit.socket
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Wed Nov 15 12:25:24 2023 from 10.10.10.4
# exit
logout
Connection to 10.20.10.4 closed.
```

For example, to access the proxy server with IP 10.30.10.4, run:

```
# ssh -A -o ServerAliveInterval=60 -o ServerAliveCountMax=600 -o ProxyCommand="ssh -W %h:%p root@159.23.96.129" root@10.30.10.4
The authenticity of host '10.30.10.4 (<no hostip for proxy command>)' can't be e
stablished.
ECDSA key fingerprint is SHA256:mY+dkcNTBIZeb/c0JnKBf+/ipaPUBapXF14dnWAOzDc.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.30.10.4' (ECDSA) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket
```

```
Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Nov 9 11:58:39 2023 from 10.10.10.4
```

You can ping the gateways to confirm connectivity.

```
[root@fsyj-private-svs-1 ~]# ping 10.51.0.1
PING 10.51.0.1 (10.51.0.1) 56(84) bytes of data.
64 bytes from 10.51.0.1: icmp_seq=1 ttl=53 time=1.76 ms
64 bytes from 10.51.0.1: icmp_seq=2 ttl=53 time=1.100 ms
64 bytes from 10.51.0.1: icmp_seq=3 ttl=53 time=1.81 ms
--- 10.51.0.1 ping statistics ---
```

```
3 packets transmitted, 3 received, 0% packet loss, time 2003ms  
rtt min/avg/max/mdev = 1.758/1.856/1.998/0.102 ms
```

```
[root@fsyj-private-svs-1 ~]# ping 10.52.0.1  
PING 10.52.0.1 (10.52.0.1) 56(84) bytes of data.  
64 bytes from 10.52.0.1: icmp_seq=1 ttl=53 time=1.79 ms  
64 bytes from 10.52.0.1: icmp_seq=2 ttl=53 time=1.86 ms  
64 bytes from 10.52.0.1: icmp_seq=3 ttl=53 time=2.04 ms  
--- 10.52.0.1 ping statistics ---  
9 packets transmitted, 9 received, 0% packet loss, time 8016ms  
rtt min/avg/max/mdev = 1.753/1.802/1.867/0.057 ms
```

## Use deployable architecture workspace for the FalconStor StorSafe tile

Once your workspace is ready, you can run the FalconStor StorSafe tile and just select that workspace for your CRN. Then, enter the two network names and the SSH key name created in that workspace. Follow your deployment according to steps provided earlier in this document.

A value for each of the following parameters is required. A default value might be set for some parameters. You can choose to accept the default value or update it.

| Parameter                   | Description                                                                                                                                             | Value                                                                                                                                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| crn                         | Power Systems Virtual Server CRN                                                                                                                        | Select a value<br>FalconStor - Dal12 (dal12)<br>FalconStor - Lon06 (lon06)<br>FalconStor Hybrid Cloud (dal12)<br>fs-draas-tok04-power-workspace (to...<br>fsyj-syd04-power-workspace (syd04) |
| instance_name               | The name to assign to the VTL instance                                                                                                                  |                                                                                                                                                                                              |
| license_repository_capacity | The VTL licensed repository capacity in terabytes                                                                                                       |                                                                                                                                                                                              |
| memory                      | The amount of memory to assign to the VTL in gigabytes. Use the following formula: $\text{memory} \geq 16 + (2 * \text{license\_repository\_capacity})$ | 16                                                                                                                                                                                           |
| network_1                   | The first network ID or name to assign to the VTL instance, as defined for the selected Power Systems Virtual Server CRN                                | Enter network_1                                                                                                                                                                              |
| network_2                   | The second network ID or name to assign to the VTL instance, as defined for the selected Power Systems Virtual Server CRN                               | Enter network_2                                                                                                                                                                              |

**Summary**

**FalconStor StorSafe VTL for PowerVS Cloud**  
 Deployment target: Power Systems Virtual Server  
 Delivery method: Server Image  
 Workspace: vtile-tags-v10-03-0-11-09-2023  
 Resource group: Default  
 Location: Frankfurt

**Information required:** Set the required input variables to install FalconStor StorSafe VTL for PowerVS Cloud.

|                             |                                                                                                                                                         |                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| crn                         | Power Systems Virtual Server CRN                                                                                                                        | fsyj-syd04-power-workspace (syd04) |
| instance_name               | The name to assign to the VTL instance                                                                                                                  | fsyj-dp-vtl-instance               |
| license_repository_capacity | The VTL licensed repository capacity in terabytes                                                                                                       | 1                                  |
| memory                      | The amount of memory to assign to the VTL in gigabytes. Use the following formula: $\text{memory} \geq 16 + (2 * \text{license\_repository\_capacity})$ | 16                                 |
| network_1                   | The first network ID or name to assign to the VTL instance, as defined for the selected Power Systems Virtual Server CRN                                | bkp_net                            |
| network_2                   | The second network ID or name to assign to the VTL instance, as defined for the selected Power Systems Virtual Server CRN                               | mgmt_net                           |
| network_3                   | The third network ID or name to assign to the VTL instance, as defined for the selected Power Systems Virtual Server CRN                                | mgmt_net                           |
| processors                  | The number of vCPUs to assign to the VTL as visible within the guest Operating System                                                                   | 1                                  |
| ssh_key_name                | The name of the public SSH RSA key to access the VTL instance, as defined for the selected Power Systems Virtual Server CRN                             | fsyj-syd04-pcs-ssh-key             |
| storage_type                | The type of storage tier to assign for storage volume performance: 'tier1' or 'tier3'                                                                   | tier1                              |
| sys_type                    | The type of system on which to create the VTL: 's922', 'e980'                                                                                           | s922                               |

**Summary**

**FalconStor StorSafe VTL for PowerVS Cloud**  
 Deployment target: Power Systems Virtual Server  
 Delivery method: Server Image  
 Workspace: vtile-tags-v10-03-0-11-14-2023  
 Resource group: Default  
 Location: Frankfurt

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 LICENSE

**Install**

Once FalconStor StorSafe tile deployment is completed, you can see the related virtual appliance instance in the workspace.

Power Systems Virtual Server
Workspaces
fsyj-syd04-power-workspace
Compute
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Virtual appliances
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CRN

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Appliance name

Appliance type

IP address

Status

|                      |     |                                     |        |
|----------------------|-----|-------------------------------------|--------|
| fsyj-dp-vtl-instance | VTL | 10.52.0.76, 10.51.0.140, 10.51.0.78 | Active |
|----------------------|-----|-------------------------------------|--------|

Create from catalog

Run the SSH command to connect to FalconStor StorSafe with the *centos* user account from the jump box:

```
ssh -A -o ServerAliveInterval=60 -o ServerAliveCountMax=600 -o
ProxyCommand="ssh -W %h:%p root@159.23.96.129" centos@10.52.0.76
```

```
#####
##                                ##
##          AUTHORIZED USERS ONLY          ##
##                                ##
##          The information on this computer is protected by          ##
##          intellectual property rights. You activity may be          ##
##          monitored and recorded.          ##
##                                ##
#####
```

```
#####
##                                ##
##          AUTHORIZED USERS ONLY          ##
##                                ##
##          The information on this computer is protected by          ##
##          intellectual property rights. You activity may be          ##
##          monitored and recorded.          ##
##                                ##
#####
```

=====

IBM Service and Productivity Tools for Linux on Power

IBM value-added software for Linux on Power servers  
is available to be installed.

You can set up IBM remote repositories at any time  
by running as root:

```
# /opt/ibm/lop/configure
```

=====

```
[centos@fsyj-dp-vtl-instance ~]$
```